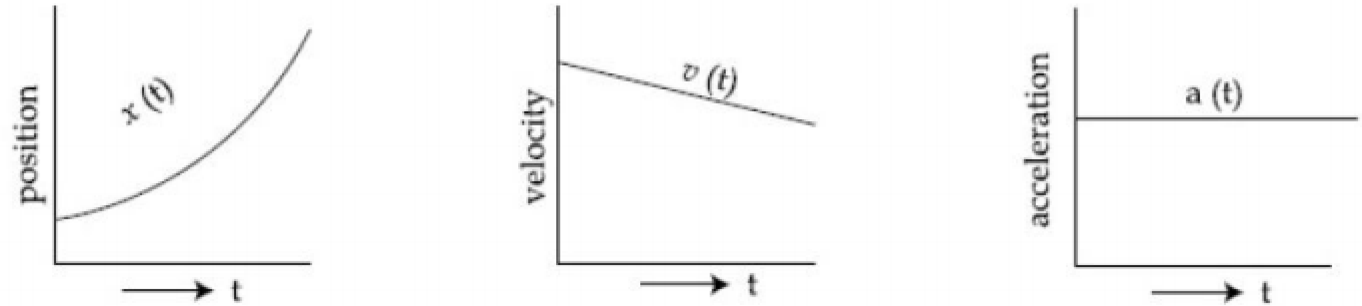
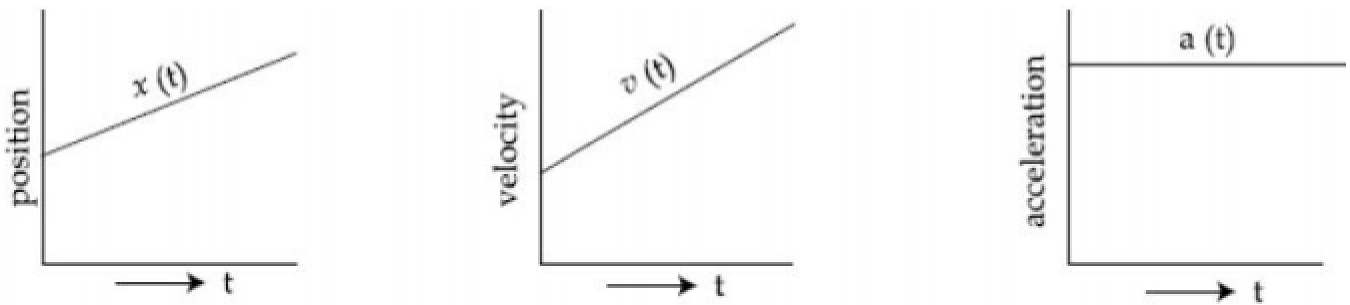
Q1: The position, velocity and acceleration of a particle moving with a constant acceleration can be represented by:

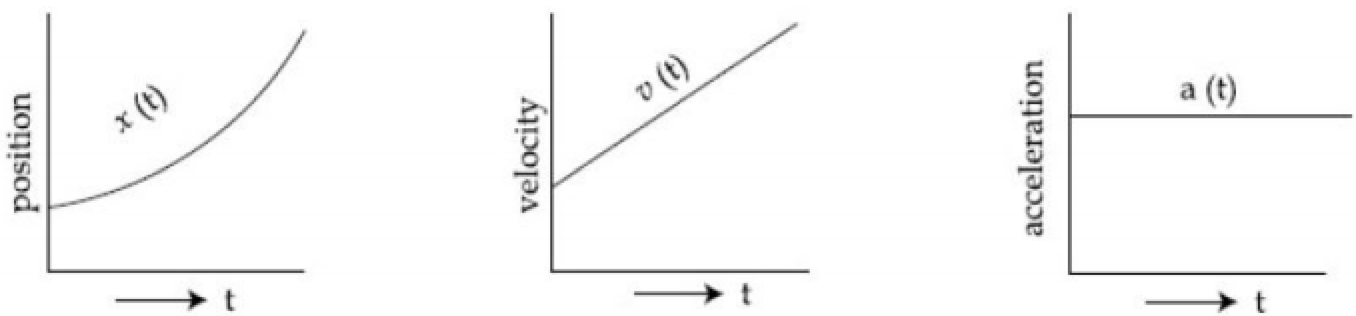
(A)



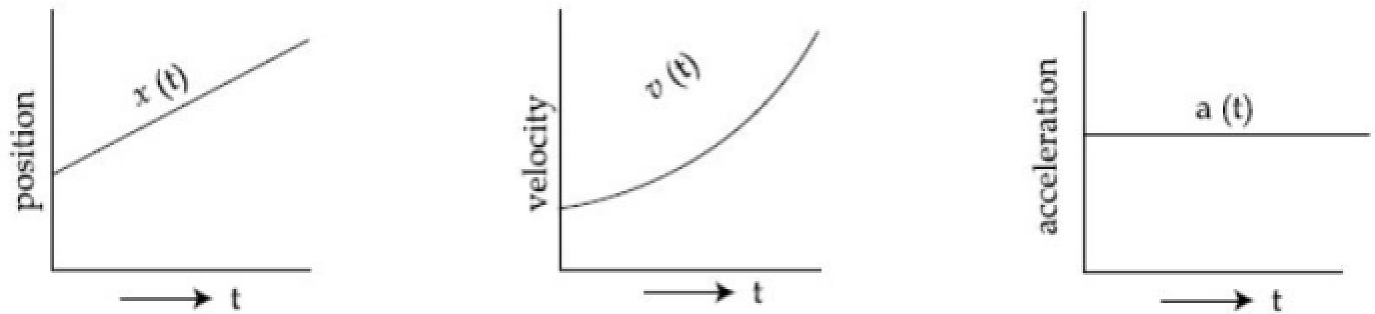
(B)



(C)



(D)



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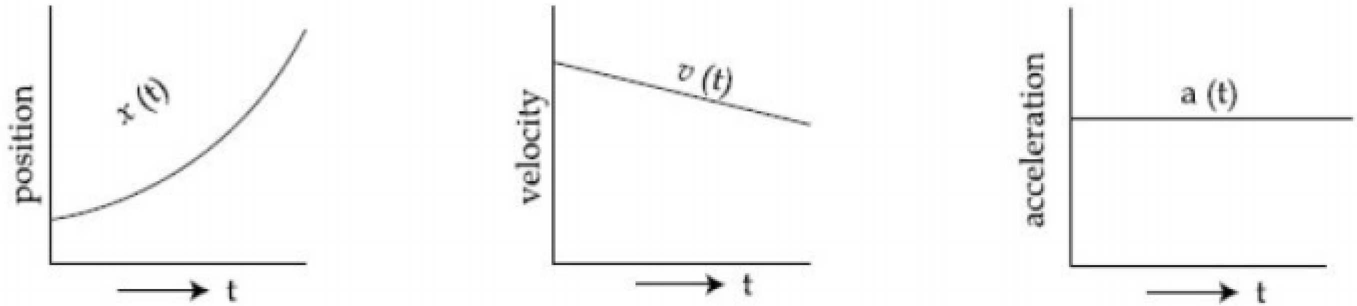

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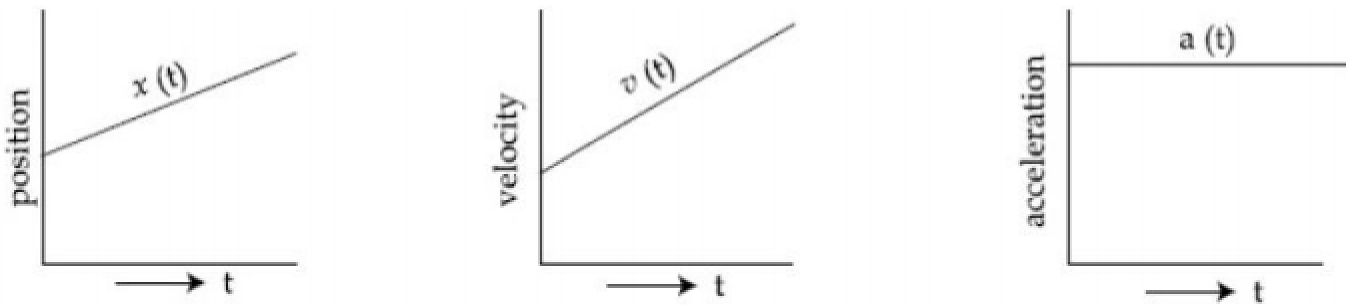
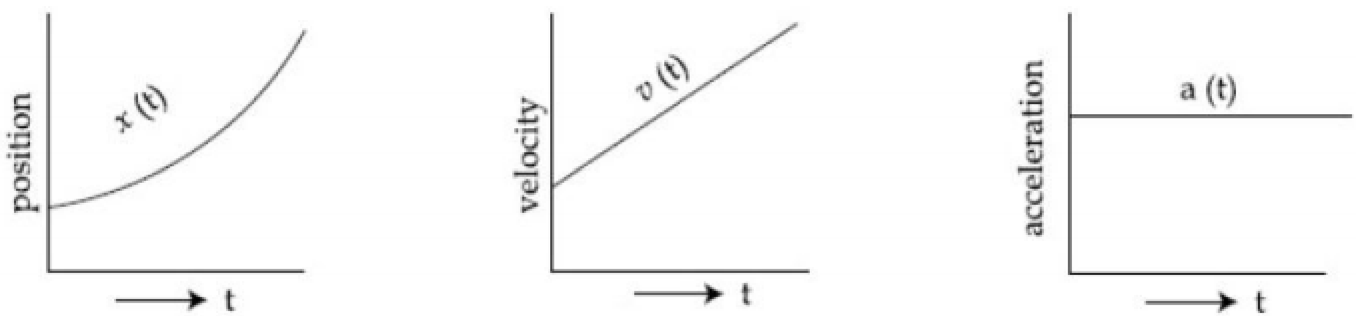
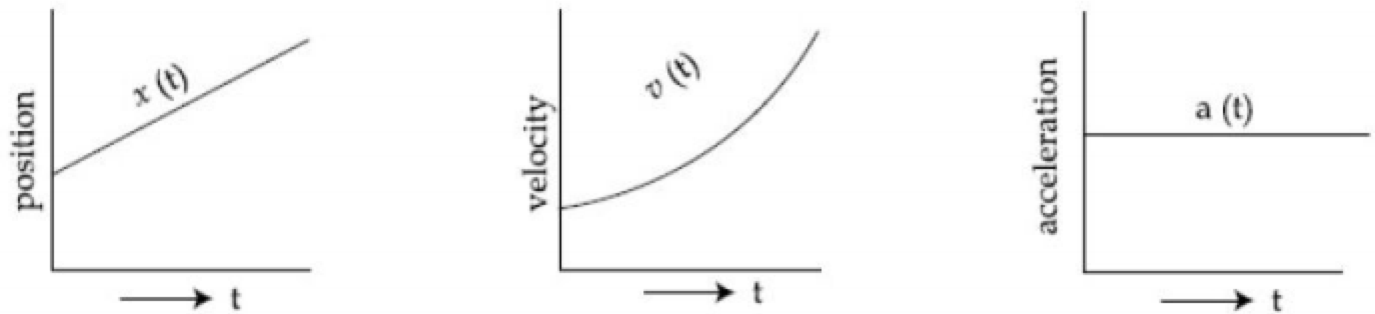


(C)



(D)



Solution:

Option (c) represent correct graph for particle moving with constant acceleration, as for constant acceleration velocity time graph is straight line with positive slope and x-t graph should be an opening upward parabola.

Q2: In a series LCR resonance circuit, if we change the resistance only, from a lower to higher value:

(A) The quality factor will increase

(B) The resonance frequency will increase

(C) The bandwidth of resonance circuit will increase

(D) The quality factor and the resonance frequency will remain constant

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Solution:

$$\text{For series LCR circuit}$$
$$\Delta\omega = R/L$$
$$\text{Bandwidth} \propto \frac{\text{R}}{L}$$

Bandwidth $\propto$ R

So, bandwidth will increase.

Q3: The constant power delivering machine has towed a box, which was initially at rest, along a horizontal straight line. The distance moved by the box in time ‘t’ is proportional to:

(A) $t^{1/2}$

(B) $t^{2/3}$

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(D) $t^{3/2}$

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(C) t

(D) $t^{3/2}$

Solution:

Let C be any constant.

$P = C$

$\overrightarrow{F}\,.\,\overrightarrow{v} = C$

$M\frac{dv}{dt}v = C$

$\frac{v^2}{2} \propto t$

$v \propto t^{1/2}$

$\frac{dx}{dt} \propto t^{1/2}$

$x \propto t^{3/2}$

Q4: The time period of a satellite in a circular orbit of radius R is T. The period of another satellite in a circular orbit of radius 9R is:

(A) 3 T

(B) 9 T

(C) 27 T

(D) 12 T

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(A) 3 T

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(C) 27 T

(D) 12 T

Solution:

$$T^2 \propto R^3$$

$$\left(\frac{T_1}{T}\right)^2 = \left(\frac{9R}{R}\right)^3$$

$$T_1{}^2 = T^2 \times 9^3$$

$$T_1 = T \times 3^3$$

$$T_1 = 27\,T$$

Q5: In Young's double slit arrangement, slits are separated by a gap of 0.5 mm, and the screen is placed at a distance of 0.5 m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 Å is:

(A) $5890 \times 10^{-7}\ m$

(B) $1178 \times 10^{-9}\ m$

(C) $1178 \times 10^{-6}\ m$

(D) $1178 \times 10^{-12}\ m$

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(C) $1178 \times 10^{-6}\ m$

(D) $1178 \times 10^{-12}\ m$

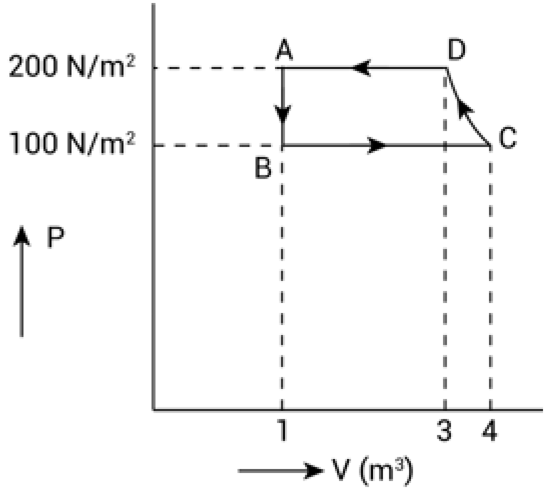
Solution:

$$\beta = \frac{\lambda D}{d} = \frac{5890 \times 10^{-10} \times 0.5}{0.5 \times 10^{-3}}$$

$$= 589 \times 10^{-6}\ \mathrm{m}$$

Distance between first and third bright fringe is $2\beta = 2 \times 589 \times 10^{-6}\ \mathrm{m} = 1178 \times 10^{-6}\mathrm{m}$

Q6: The P-V diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process CD is (use $\gamma = 1.4$):



(A) – 400 J

(B) 400 J

(C) 200 J

(D) – 500 J

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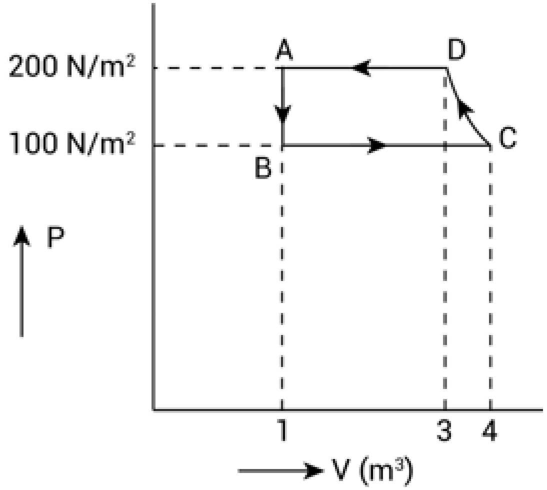

(A) – 400 J

(B) 400 J

(C) 200 J

(D) – 500 J

Solution:

Adiabatic process is from C to D

$$WD=\frac{P_2\ V_2-P_1\ V_1}{1-\gamma}$$

$$=\frac{P_D V_D-P_C V_C}{1-\gamma}$$

$$=\frac{200(3)-(100)(4)}{1-1.4}=-500\ \mathrm{J}$$

Q7: A loop of flexible wire of irregular shape carrying current is placed in an external magnetic field. Identify the effect of the field on the wire.

(A) shape of the loop remains unchanged

(B) wire gets stretched to become straight

(C) loop assumes circular shape with its plane parallel to the field

(D) loop assumes circular shape with its plane normal to the field

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Solution:

Every part (dl) of the wire is pulled by force $|F| = i\,(dl)\,B$ acting perpendicular to current & magnetic field giving it a shape of circle.

Q8: A thin circular ring of mass M and radius r is rotating about its axis with an angular speed ω. Two particles having mass m each are now attached at diametrically opposite points. The angular speed of the ring will become:

(A) $\left(\frac{M+2m}{M}\right)\omega$

(B) $\left(\frac{M}{M+2m}\right)\omega$

(C) $\left(\frac{M}{M+m}\right)\omega$

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(C) $\left(\frac{M}{M+m}\right)\omega$

(D) $\left(\frac{M-2m}{M+2m}\right)\omega$

Solution:

Using conservation of angular momentum

$$\left(Mr^2\right)\omega = \left(Mr^2 + 2mr^2\right)\omega'$$

$$\omega' = \frac{M\omega}{M+2\ m}$$

Q9: In the experiment of Ohm’s law, a potential difference of 5.0 V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is:

(A) 8.4

(B) 7.5

(C) 3.9

(D) 3.0

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(A) 8.4

(B) 7.5

(C) 3.9

(D) 3.0

Solution:

$$R = \frac{\rho \ell}{A} = \frac{V}{I}$$

$$\rho = \frac{AV}{I\ell} = \frac{\pi d^2 V}{4I\ell} \qquad \therefore \quad \left(A = \frac{\pi d^2}{4}\right)$$

$$\therefore \frac{\Delta\rho}{\rho} = \frac{2\Delta d}{d} + \frac{\Delta V}{V} + \frac{\Delta I}{I} + \frac{\Delta\ell}{\ell}$$

$$\frac{\Delta\rho}{\rho} = 2\left(\frac{0.01}{5.00}\right) + \frac{0.1}{5.0} + \frac{0.01}{2.00} + \frac{0.1}{10.0}$$

$$\frac{\Delta\rho}{\rho} = 0.004 + 0.02 + 0.005 + 0.01$$

$$\frac{\Delta\rho}{\rho} = 0.039$$

$$\%\ \text{error} = \frac{\Delta\rho}{\rho} \times 100 = 0.039 \times 100 = 3.90\%$$

Q10: What will be the average value of energy along on degree of freedom for an ideal gas in thermal equilibrium at a temperature T? (k_B is Boltzmann constant)

(A) $\frac{2}{3}\, k_B T$

(B) $k_B T$

(C) $\frac{1}{2}\, k_B T$

(D) $\frac{3}{2}\, k_B T$

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(B) $k_B T$

(C) $\frac{1}{2} k_B T$

(D) $\frac{3}{2} k_B T$

Solution:

From the law of equipartition of energy, it states that the energy associated with each degree of freedom is $\frac{1}{2} k_B T$

Q11: Match List-I with List-II

List-I	**List-II**
(A) 10 km height over earth's surface	(i) Thermosphere
(B) 70 km height over earth's surface	(ii) Mesosphere
(C) 180 km height over earth's surface	(iii) Stratosphere
(D) 270 km height over earth's surface	(iv) Troposphere

(A) (A) – (ii), (B) – (i), (C) – (iv), (D) – (iii)

(B) (A) – (i), (B) – (iv), (C) – (iii), (D) – (ii)

(C) (A) – (iii), (B) – (ii), (C) – (i), (D) – (iv)

(D) (A) – (iv), (B) – (iii), (C) – (ii), (D) – (i)

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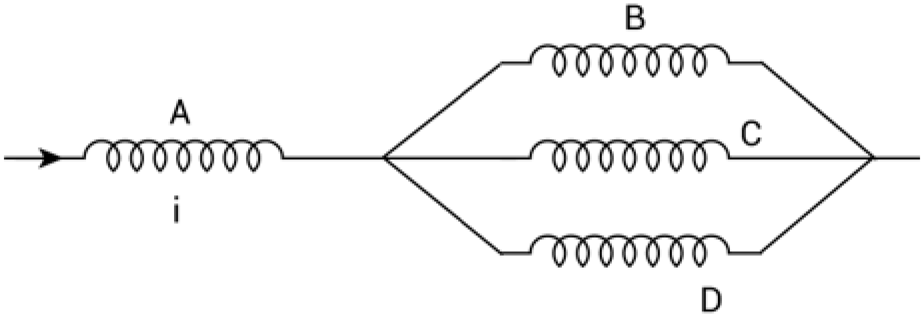
(C) (A) – (iii), (B) – (ii), (C) – (i), (D) – (iv)

(D) (A) – (iv), (B) – (iii), (C) – (ii), (D) – (i)

Solution:

Order of atmosphere stratification from bottom Troposphere, stratosphere, Mesosphere, Thermosphere
(a) $\rightarrow$ (iv)
(b) $\rightarrow$ (iii)
(c) $\rightarrow$ (ii)
(d) $\rightarrow$ (i)

Q12: Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the centre of A is 3 T, the field at the center of C would be: (Assume that the magnetic field is confined with in the volume of respective solenoid).



(A) 1 T

(B) 12 T

(C) 9 T

(D) 6 T

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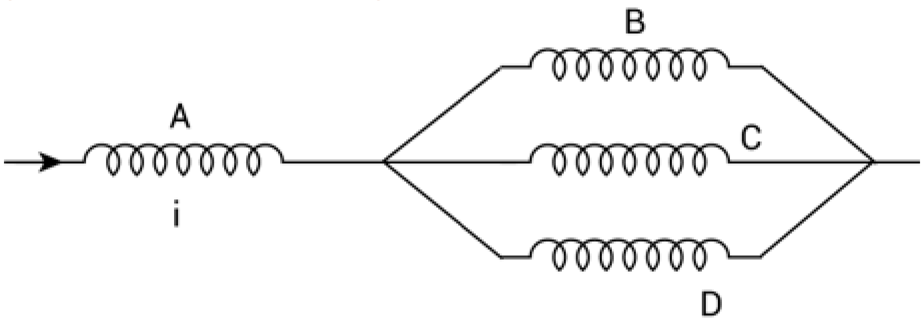

(A) 1 T

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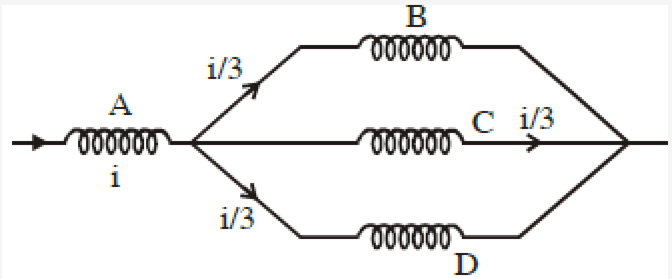
Solution:



$$B \propto i$$

$$B_A \propto i \ldots\ldots\ldots\ldots\ldots (1)$$

$$B_C \propto \frac{i}{3} \ldots\ldots\ldots\ldots (2)$$

$$\text{From } (1) \text{ and } (2)$$

$$\frac{B_A}{B_B} = 3$$

$$B_B = \frac{B_A}{3} = \frac{3}{3} = 1\,T$$

So, field at center of C **1** T.

Q13: A particle is travelling 4 times as fast as an electron. Assuming the ratio of de-Broglie wavelength of a particle to that of electron is 2:1, the mass of the particle is:

(A) $\frac{1}{8}$ times the mass of e^-

(B) 8 times the mass of e^-

(C) $\frac{1}{16}$ times the mass of e^-

(D) 16 times the mass of e^-

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(D) 16 times the mass of e^-

Solution:

$$\lambda = \frac{h}{p}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{p_e}{P_p} = \frac{m_e v_e}{m_p v_p}$$

$$2 = \frac{m_e}{m_p}\left(\frac{v_e}{4v_e}\right)$$

$$\therefore m_p = \frac{m_e}{8}$$

Q14: An AC source rated $220\ V,\ 50\ Hz$ is connected to a resistor. The time taken by the current to change from its maximum to the rms value is:

(A) 25 ms

(B) 2.5 ms

(C) 0.25 ms

(D) 2.5 s

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(D) 2.5 s

Solution:

$$i = i_0 \cos(\omega t)$$
$$i = i_0\ at\ t = 0$$
$$i = \frac{i_0}{\sqrt{2}}\ at\ \omega t = \frac{\pi}{4}$$
$$t = \frac{\pi}{4\omega} = \frac{\pi}{4(2\pi f)} = \frac{1}{8f}$$
$$t = \frac{1}{400} = 2.5\ \mathrm{ms}$$

Q15: An oil drop of radius 2 mm with a density $3\,g\,cm^{-3}$ is held stationary under a constant electric field $3.55\ \times\ 10^5\ V\ m^{-1}$ in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will possess?
Consider g = 9.81 m/s^2

(A) $48.8\ \times\ 10^{11}$

(B) $1.73\ \times\ 10^{12}$

(C) $1.73\ \times\ 10^{10}$

(D) $17.3\ \times\ 10^{10}$

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(A) 48.8×10^{11}

(B) 1.73×10^{12}

(C) 1.73×10^{10}

(D) 17.3×10^{10}

Solution:

$$qE = Mg$$
$$neE = \rho\left(\frac{4}{3}\pi r^3\right) \times g$$
$$n \times 1.6 \times 10^{-19} \times 3.55 \times 10^5 = 3 \times 10^3 \times \frac{4}{3} \times \pi \times \left(2 \times 10^{-3}\right)^3 \times 9.81$$
$$n = 173.5 \times 10^{(3-9-5+19)}$$
$$n = 1.73 \times 10^{10}$$

Q16: A plane electromagnetic wave of frequency 100 MHz is travelling in vacuum along the x-direction. At a particular point in space and time, $\vec{B} = 2.0 \times 10^{-8}\ \hat{k}\ T$. (where, $\hat{k}$ is unit vector along z-direction) What is $\vec{E}$ at this point?
(speed of light $c = 3 \times 10^{8}\ m/s$)

(A) $0.6\ \hat{k}\ V/m$

(B) $0.6\ \hat{j}\ V/m$

(C) $6.0\ \hat{j}\ V/m$

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(D) $6.0\ \hat{k}\ V/m$

Solution:

$\Rightarrow f = 100\, MHz$
We know that
$|E| = c\,|B|$
As wave is going in X-direction and magnetic field is along Z-direction so from
$\vec{S} = \vec{E} \times \vec{B}$
We get direction of E along Y-direction
$|E| = 3 \times 10^{8} \times 2 \times 10^{-8}$
$E = 6V/m$
$\vec{E} = (6V/m)\ j$

Q17: Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionization potential of this hydrogen atom will be: (Assume, Nucleus is at rest)

(A) 13.6 eV

(B) 27.2 eV

(C) 331.2 eV

(D) 2815.2 eV

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Solution:

$$\mathrm{E} \propto \frac{1}{\mathrm{r}} \quad \mathrm{r} \propto \frac{1}{\mathrm{m}}$$

$$\mathrm{E} \propto \mathrm{m}$$

$$\text{Ionization potential } = 13.6 \times \frac{(\text{Mass }_{\mu})\mathrm{eV}}{(\text{Mass }_{e})}$$

$$= 13.6 \times 207\mathrm{eV} = 2815.2 \text{ eV}$$

Q18: A radioactive sample disintegrates via two independent decay processes having half-lives $T_{1/2}^{(1)}$ and $T_{1/2}^{(2)}$ respectively. The effective half-life, $T_{1/2}$ of the nuclei is :

(A) $T_{1/2} = T_{1/2}^{(1)} + T_{1/2}^{(2)}$

(B) $T_{1/2} = \frac{T_{1/2}^{(1)} + T_{1/2}^{(2)}}{T_{1/2}^{(1)} - T_{1/2}^{(2)}}$

(C) $T_{1/2} = \frac{T_{1/2}^{(1)} \cdot T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$

(D) None of the above

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(C) $T_{1/2} = \frac{T_{1/2}^{(1)} \cdot T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$

(D) None of the above

Solution:

$$\lambda_{eq} = \lambda_1 + \lambda_2$$

$$\frac{1}{T_{1/2}} = \frac{1}{T_{1/2}^{(1)}} + \frac{1}{T_{1/2}^{(2)}}$$

$$T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$$

Q19: Your friend is having eyesight problem. She is not able to see clearly a distant uniform window and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as:

(A) Presbyopia with Astigmatism

(B) Astigmatism

(C) Myopia with Astigmatism

(D) Myopia and hypermetropia

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(C) Myopia with Astigmatism

(D) Myopia and hypermetropia

Solution:

Near-sightedness (myopia) is a common vision condition in which you can see objects near to you clearly, but objects farther away are blurry.
Astigmatism is a type of refractive error in which the eye does not focus light evenly on the retina. This results in distorted or blurred vision at any distance.
So, he is suffering from both eye defects, astigmatism, and myopia.

Q20: The time period of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$. The measured value of the length of pendulum is 10 cm known to a 1 mm accuracy. The time for 200 oscillations of the pendulum is found to be 100 second using a clock of 1 s resolution. The percentage accuracy in the determination of ‘g’ using this pendulum is ‘x’. The value of ‘x’ to the nearest integer is,

(A) 4%

(B) 5%

(C) 3%

(D) 2%

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(B) 5%

(C) 3%

(D) 2%

Solution:

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + 2\frac{\Delta T}{T} = \frac{0.1}{10} + 2\left(\frac{\frac{1}{200}}{0.5}\right)$$

$$\frac{\Delta g}{g} = \frac{1}{100} + \frac{1}{50}$$

$$\frac{\Delta g}{g} \times 100 = 3\%$$

Q21: An npn transistor operates as a common emitter amplifier with a power gain of 10^6. The input circuit resistance is $100\ \Omega$ and the output load resistance is $10\ k\Omega$. The common emitter current gain $'\beta'$ will be .... (Round off to the Nearest Integer)

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100.00

Solution:

$$10^6 = \beta^2 \times \frac{R_0}{R_i}$$

$$10^6 = \beta^2 \times \frac{10^4}{10^2}$$

$$\beta^2 = 10^4 \Rightarrow \beta = 100$$

Q22: A person is swimming with a speed of 10 m/s at an angle of 120º with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is x(in m/s). The value of 'x' to the nearest integer is ....

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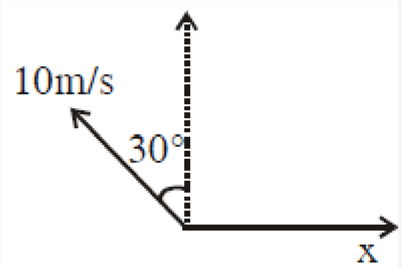
005.00

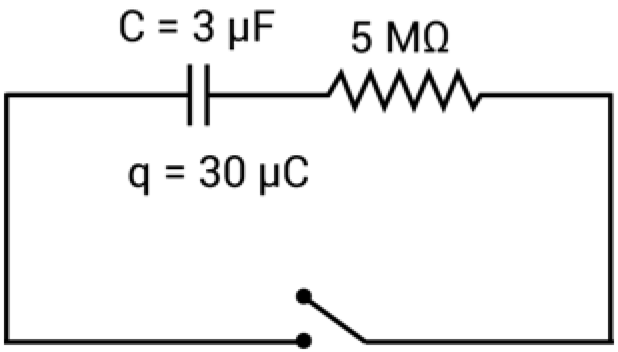
Solution:



$x = 10\ sin\ 30°$

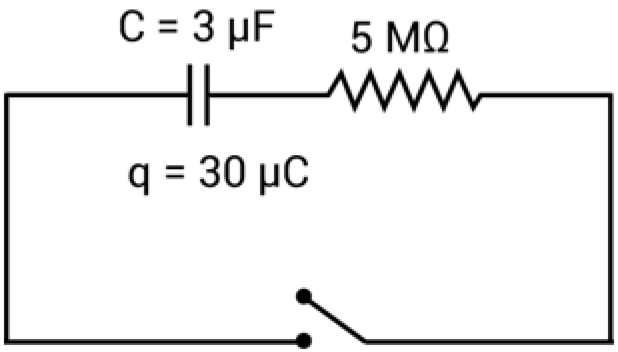
$x\ =\ 5\ m/s$

Q23:



The circuit shown in the figure consists of a charged capacitor of capacity $3\ \mu F$ and a charge of $30\ \mu C.$ At time t = 0, when the key is closed, the value of current flowing through the $5\ M\Omega$ resistor is 'x' $\mu A.$

The value of 'x' to the nearest integer is ....

Q23: 

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The value of 'x' to the nearest integer is ....

002.00

Solution:

Charging of capacitor is given by

$Q(t) = Q_0\left(1 - e^{-t/Rc}\right)$

Differentiating both side w.r.t time

$I(t) = \frac{Q_0}{RC} e^{-t/RC}$

At $t = 0$

$I(0) = \frac{Q_0}{RC}$

Given $Q_0 = 30\mu C$

$R = 5M\Omega$

$C = 3\mu F$

$RC = (3 \times 5)\, s = 15s$

$I(0) = I_0 = \frac{30\mu C}{15s} = 2\mu A$

Q24: A bullet of mass 0.1 kg is fired on a wooden block to pierce through it, but it stops after moving a distance of 50 cm into it. If the velocity of bullet before hitting the wood is 10 m/s and it slows down with uniform deceleration, then the magnitude of effective retarding force on the bullet is 'x' N.

The value of 'x' to the nearest integer is ....

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The value of 'x' to the nearest integer is ....

010.00

Solution:

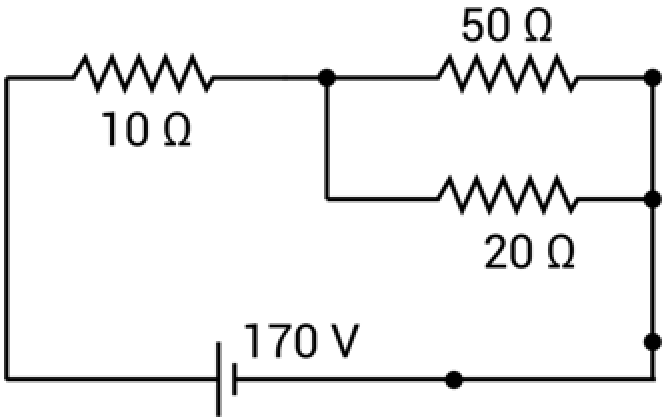
$$v^2 = u^2 + 2as$$

$$0 = (10)^2 + 2(-a)\left(\frac{1}{2}\right)$$

$$a = 100\ m/s^2$$

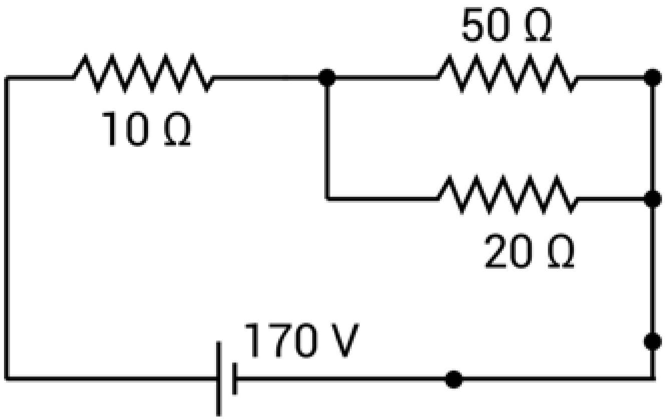
$$F = ma = (0.1)(100) = 10\ N$$

Q25: The voltage across the $10\ \Omega$ resistor in the given circuit is x volt.



The value of 'x' to the nearest integer is ....

Q25: The voltage across the $10\ \Omega$ resistor in the given circuit is x volt.

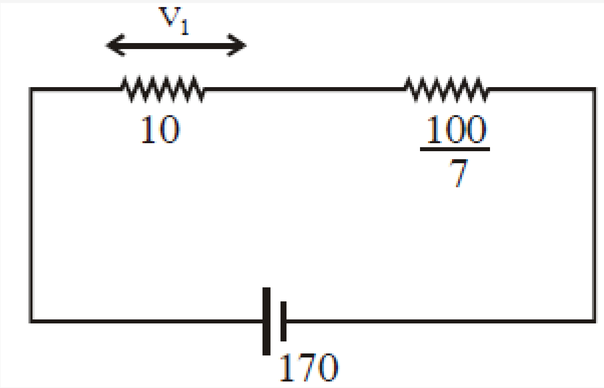


The value of 'x' to the nearest integer is ....

070.00

Solution:

$$R_{eq_1} = \frac{50 \times 20}{70} = \frac{100}{7}$$



$$R_{eq} = \frac{170}{7}$$

$$v_1 = \left[\frac{170}{\frac{170}{7}}\right] \times 10 = 70\text{V}$$

Q26: A parallel plate capacitor has plate area 100 m^2 and plate separation of 10 m. The space between the plates is filled up to a thickness 5 m with a material of dielectric constant of 10. The resultant capacitance of the system is.

The value of $\varepsilon_0 = 8.85 \times 10^{-12}\ F.m^{-1}$

The value of to the nearest integer is ………………. .

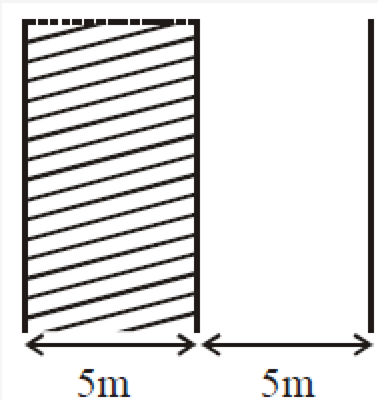
Q26: A parallel plate capacitor has plate area 100 m^2 and plate separation of 10 m. The space between the plates is filled up to a thickness 5 m with a material of dielectric constant of 10. The resultant capacitance of the system is.

The value of $\varepsilon_0 = 8.85 \times 10^{-12} F.m^{-1}$

The value of to the nearest integer is ....

161.00

Solution:



$$A = 100\ \mathrm{m}^2$$

$$\text{Using } C = \frac{K\varepsilon_0\ \mathrm{A}}{\mathrm{d}}$$

$$\mathrm{C}_1 = \frac{10\varepsilon_0(100)}{5}$$

$$= 200\varepsilon_0$$

$$\mathrm{C}_2 = \frac{\varepsilon_0(100)}{5} = 20\varepsilon_0$$

$$\mathrm{C}_1 \text{ and } \mathrm{C}_2 \text{ are in series so } \mathrm{C}_{\text{eqv.}} = \frac{\mathrm{C}_1\mathrm{C}_2}{\mathrm{C}_1 + \mathrm{C}_2}$$

$$= \frac{4000\varepsilon_0}{220}$$

$$= 160.9 \times 10^{-12} \simeq 161\,\mathrm{pF}$$

Q27: A particle performs simple harmonic motion with a period of 2 second. The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}\ s.$

The value of ‘a’ to the nearest integer is ………………. .

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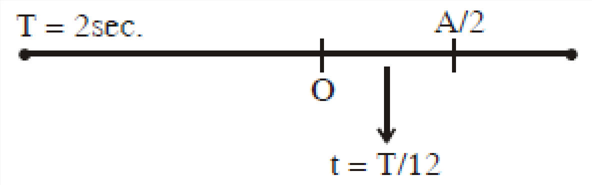
The value of 'a' to the nearest integer is ………………… .

006.00

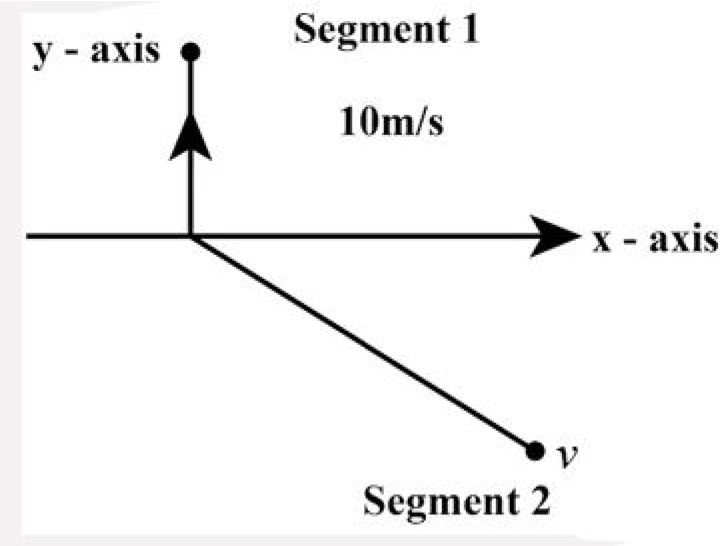
Solution:



$t = \frac{2}{12} = \frac{1}{6}$

Q28: A ball of mass 10 kg moving with a velocity $10\sqrt{3}$m/s along the x-axis, hits another ball of mass 20 kg which is at rest. After the collision, first ball comes to rest while the second ball disintegrates into two equal pieces. One piece starts moving along y-axis with a speed of 10 m/s. The second piece starts moving at an angle of 30º with respect to the x-axis. The velocity of the ball moving at 30º with x-axis is $x\ m/s$. The configuration of pieces after collision is shown in the figure below.

The value of x to the nearest integer is ....



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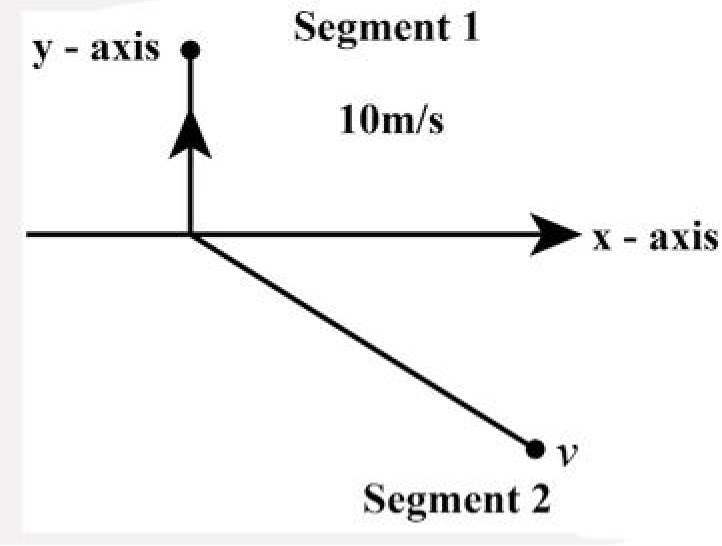
The value of x to the nearest integer is ....



020.00

Solution:

Let velocity of 2nd fragment is $\vec{v}$ then by conservation of linear momentum

$$10(10\sqrt{3})\hat{i} = (10)(10\hat{j}) + 10\vec{v}$$

$$\Rightarrow \vec{v} = 10\sqrt{3}\hat{i} - 10\hat{j}$$

$$|\vec{v}| = \sqrt{300 + 100} = \sqrt{400} = 20 \text{ m/s}$$

Q29: Two separate wires A and B are stretched by 2 mm and 4 mm respectively, when they are subjected to a force of 2 N. Assume that both the wires are made up of same material and the radius of wire B is 4 times that of the radius of wire A. The length of the wires A and B are in the ratio of a:b. Then $\frac{a}{b}$ can be expressed as $\frac{1}{x}$ where x is ....

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032.00

Solution:

$$\text{For A,} \frac{F}{\pi r^2} = Y\frac{2\text{ mm}}{\text{a}} \ldots\ldots(1)$$

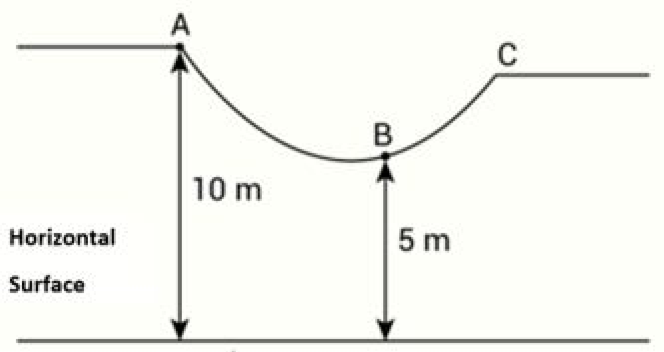
$$\text{For B,} \frac{F}{\pi \cdot 16 r^2} = Y\frac{4\text{ mm}}{\text{b}} \ldots\ldots(2)$$

$$\therefore \text{Divide (1) by (2)}$$

$$16 = \frac{2\text{ b}}{4\text{a}}$$

$$\frac{a}{b} = \frac{1}{32} = 32$$

Q30: As shown in the figure, a particle of mass 10 kg is placed at a point A. When the particle is slightly displaced to its right, it starts moving and reaches the point B. The speed of the particle at B is $x\ m/s$.



$\left(Take\ g\ =\ 10\ m/s^2\right)$

The value of 'x' to the nearest integer is ....

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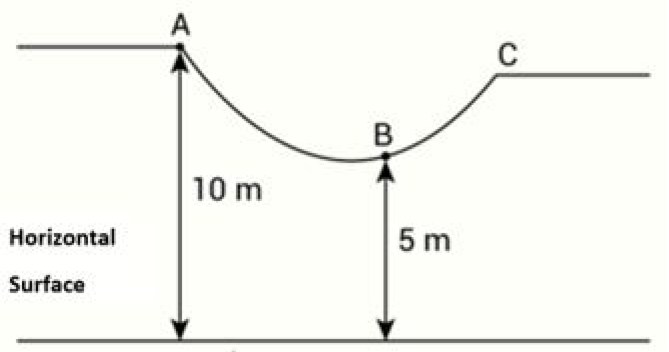

$\left(Take\ g\ =\ 10\ m/s^2\right)$

The value of 'x' to the nearest integer is ....

010.00

Solution:

Using work energy theorem,

$$W_g = \Delta K.E$$

$$(10)\,(g)\,(5) = \frac{1}{2}(10)v^2 - 0$$

$$v = 10\ \ \mathrm{m/s}$$

Q31: Match List-I with List-II

List-I	**List-II**
(A) $CaOCl_2$	**(i)** Antacid
(B) $CaSO_4 \cdot \frac{1}{2} H_2O$	**(ii)** Cement
(C) CaO	**(iii)** Bleaching powder
(D) $CaCO_3$	**(iv)** Plaster of Paris

Choose the most appropriate answer from the options given below :

(A) (A) – (iii), (B) – (ii), (C) – (i), (D) – (iv)

(B) (A) – (iii), (B) – (iv), (C) – (ii), (D) – (i)

(C) (A) – (iii), (B) – (ii), (C) – (iv), (D) – (i)

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(D) (A) – (i), (B) – (iv), (C) – (iii), (D) – (ii)

Solution:

$CaOCl_2$ → Bleaching power
$CaSO_4 \cdot \frac{1}{2} H_2O$ →Plaster of paris
CaO → cement
$CaCO_3$ → Antacid

Q32: Compound with molecular formula C_3H_6O can show :

(A) Both positional isomerism and metamerism

(B) Metamerism

(C) Functional group isomerism

(D) Positional isomerism

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(A) Both positional isomerism and metamerism

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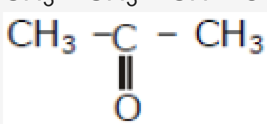
Solution:

C_3H_6O DOU = 1

$CH_3 - CH_3 - CH = O$ and



are functional isomer

Q33: A certain orbital has no angular nodes and two radial nodes. The orbital is :

(A) 2s

(B) 3s

(C) 2p

(D) 3p

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(A) 2s

(B) 3s

(C) 2p

(D) 3p

Solution:

No. of angular nodes $l = 0$
Radial nodes = $n - l - 1 = n - 0 - 1 = 2$
$n = 3$
answer is 3s

Q34: Match List-I with List-II :

List-I (Class of Drug)	**List-II (Example)**
(A) Antacid	**(i)** Novestrol
(B) Artificial Sweetener	**(ii)** Cimetidine
(C) Antifertility	**(iii)** Valium
(D) Tranquilizers	**(iv)** Alitame

Choose the most appropriate match :

(A) (A) – (iv), (B) – (iii), (C) – (i), (D) – (ii)

(B) (A) – (ii), (B) – (iv), (C) – (i), (D) – (iii)

(C) (A) – (ii), (B) – (iv), (C) – (iii), (D) – (i)

(D) (A) – (iv), (B) – (i), (C) – (oi), (D) – (iii)

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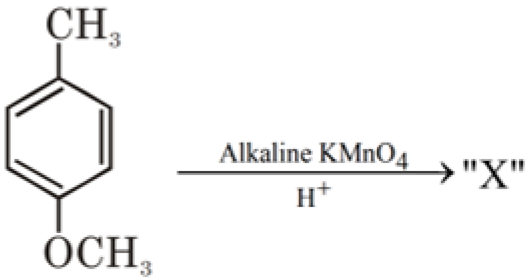
(D) (A) – (iv), (B) – (i), (C) – (oi), (D) – (iii)

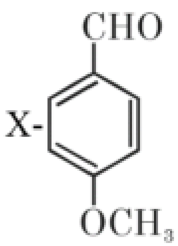
Solution:

Antacid → Cimetidine
Artificial Sweetener →Alitame
Antifertility → Novestrol
Tranquilizers → Valium

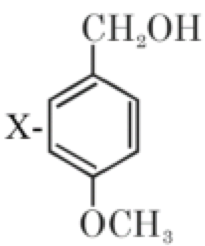
Q35: 

Considering the above chemical reaction, identify the product “X” :

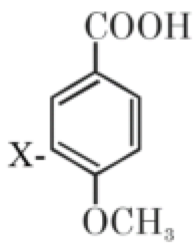
(A)



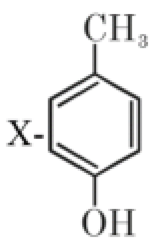
(B)

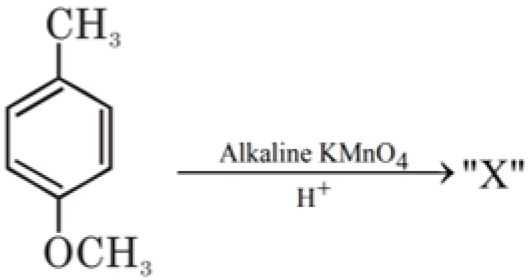


(C)



(D)



Q35: 

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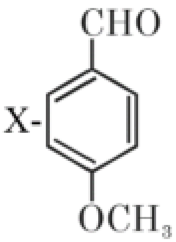
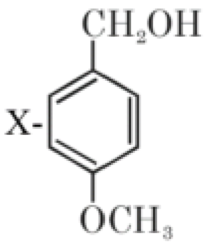
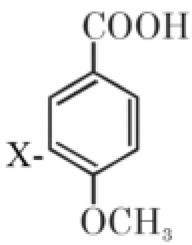


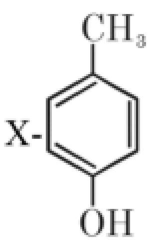
(B)



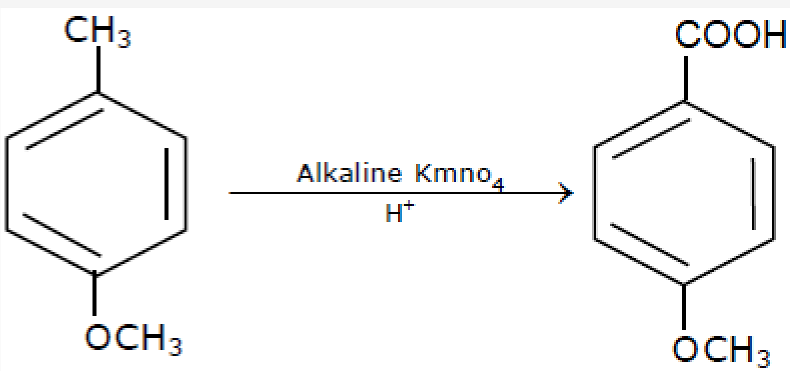
(C)



(D)



Solution:



Q36: Match List-I with List-II :

List-I (Process)	**List-II (Catalyst)**
(A) Deacon's process	**(i)** ZSM-5
(B) Contact process	**(ii)** $CuCl_2$
(C) Cracking of hydrocarbons	**(iii)** Particles 'Ni'
(D) Hydrogenation of vegetable oils	**(iv)** V_2O_5

Choose the most appropriate answer from the options given below :

(A) (A) – (iv), (b) – (ii), (C) – (i), (D) – (iii)

(B) (A) – (iii), (b) – (i), (C) – (iv), (D) – (ii)

(C) (A) – (ii), (b) – (iv), (C) – (i), (D) – (iii)

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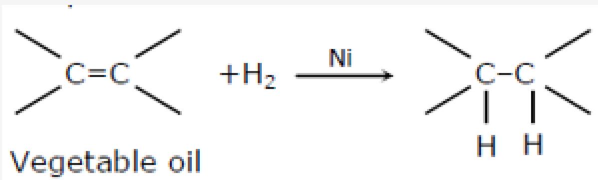
(D) (A) – (i), (b) – (iii), (C) – (ii), (D) – (iv)

Solution:

Deacon’s process is used for indusrial preparation of chlorine gas

$HCl + O_2 \xrightarrow{CuCl_2} H_2O + Cl_2$

Contact process in used for industrial preparation of sulphuric acid & V_2O_5 in catalyst involved in the process.



$$CxH_y \xrightarrow[\text{Cracking}]{\text{ZSM}-5} \text{Products}$$

Q37: A non-reducing sugar “A” hydrolyses to give two reducing mono saccharides. Sugar A is:

(A) Sucrose

(B) Glucose

(C) Fructose

(D) Galactose

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(B) Glucose

(C) Fructose

(D) Galactose

Solution:

$$\underset{\substack{\text{Sucrose}\\\text{(Non-reducing sugar)}}}{C_{12}H_{22}O_{11}} + H_2O \rightarrow \underset{\text{D-glucose}}{C_6H_{12}O_6} + \underset{\text{D-fructose}}{C_6H_{12}O_6}$$

Q38: Match List-I with List-II :

List-I (Chemicals)	**List-II (Use Preparation/Constituent)**
(A) Alcoholic potassium hydroxide	**(i)** electrodes in batteries
(B) $Pd/BaSO_4$	**(ii)** obtained by addition reaction
(C) BHC (Benzene hexachloride)	**(iii)** used for β-elimination reaction
(D) Polyacetylene	**(iv)** Lindlar's Catalyst

Choose the most appropriate match:

(A) (A) – (iii), (B) – (iv), (C) – (ii), (D) – (i)

(B) (A) – (iii), (B) – (i), (C) – (iv), (D) – (ii)

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Choose the most appropriate match:

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(D) (A) – (ii), (B) – (i), (C) – (iv), (D) – (iii)

Solution:

Alcoholic KOH ⇒ Used for β elimination reaction
$Pd/BaSO_4$ ⇒ Lindlar's catalyst
BHC (Benzene hexachloride) ⇒ Addition product of benzene and chloride
Polyacetylene ⇒ used in electrodes in batteries

Q39: Given below are two Statements: One is labelled as Assertion A and the other is labelled as Reason R :

Assertion: During the boiling of water having temporary hardness, $Mg\left(HCO_3\right)_2$ is converted to $MgCO_3$.

Reason: Temporary Hardness can be removed by Boiling.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both A and R are true but R NOT the correct explanation of A

(B) Both A and R are true and R is the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q39: Given below are two Statements: One is labelled as Assertion A and the other is labelled as Reason R :

Assertion: During the boiling of water having temporary hardness, $Mg\ (HCO_3)_2$ is converted to $MgCO_3$.

Reason: Temporary Hardness can be removed by Boiling.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both A and R are true but R NOT the correct explanation of A

(B) Both A and R are true and R is the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Solution:

$$\underset{\text{Temporary Hardness}}{Mg(HCO_3)_2} \xrightarrow{\text{Boiling}} Mg(OH)_2\downarrow + CO_2$$

Q40: The chemical that is added to reduce the melting point of the reaction mixture during the extraction of aluminium is :

(A) Kaolite

(B) Bauxite

(C) Cryolite

(D) Calamine

Q40: The chemical that is added to reduce the melting point of the reaction mixture during the extraction of aluminium is :

(A) Kaolite

(B) Bauxite

(C) Cryolite

(D) Calamine

Solution:

For reducing the melting point of Alumina, Cryolite i.e. Na_3AlF_6 is added.

Q41: The ionic radius of Na^+ ion is 1.02 Å. The ionic radii (in Å) of Mg^{2+} and Al^{3+}, respectively, are :

(A) 1.05 and 0.99

(B) 0.72 and 0.54

(C) 0.68 and 0.72

(D) 0.85 and 0.99

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(C) 0.68 and 0.72

(D) 0.85 and 0.99

Solution:

For iso-electronic system

$r \propto \frac{1}{Z_{eff}}$

$$\underbrace{Na^+ \qquad Mg^{2+} \qquad Al^{3+}}$$

$Z=11$ $\quad$ $Z=12$ $\quad$ $Z=13$ $\quad$ 10 electron in each

Z=atomic No.

Q42: The number of ionisable hydrogens present in the product obtained from a reaction of phosphorus trichloride and phosphonic acid is :

(A) 2

(B) 3

(C) 1

(D) 0

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(A) 2

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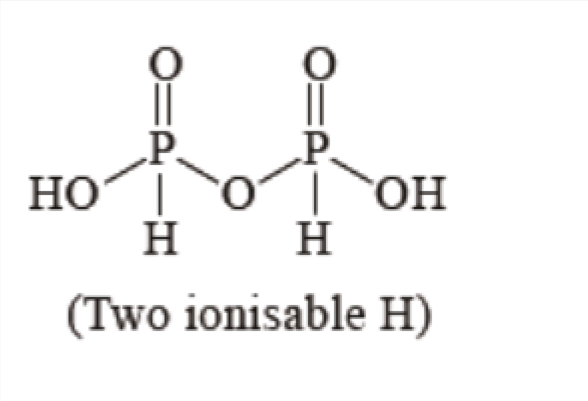
(C) 1

(D) 0

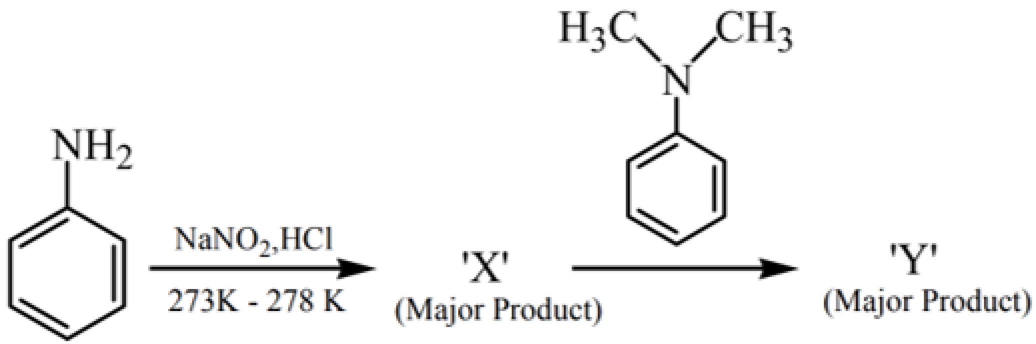
Solution:

$$PCl_3 + H_3PO_3 + H_2O \rightarrow \underset{\text{pyrophosphorous acid}}{H_4P_2O_5} + HCl$$



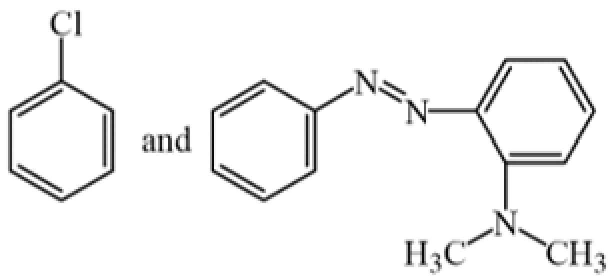
Structure of pyrophosphorous acid shows that it has two acidic or ionisable hydrogen .

Q43:

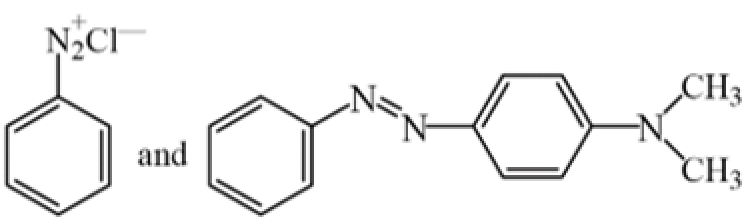


Considering the above reaction, X and Y respectively are:

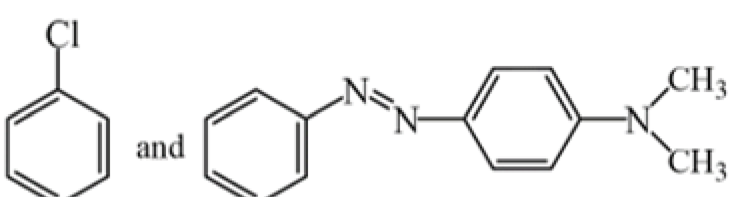
(A)

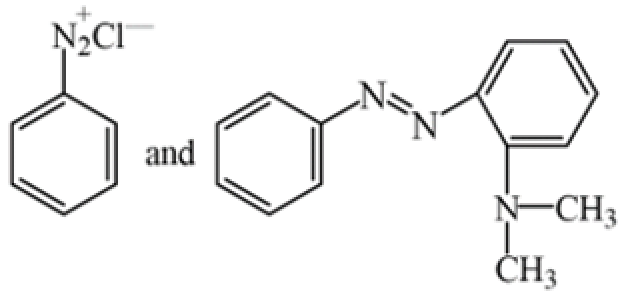


(B)

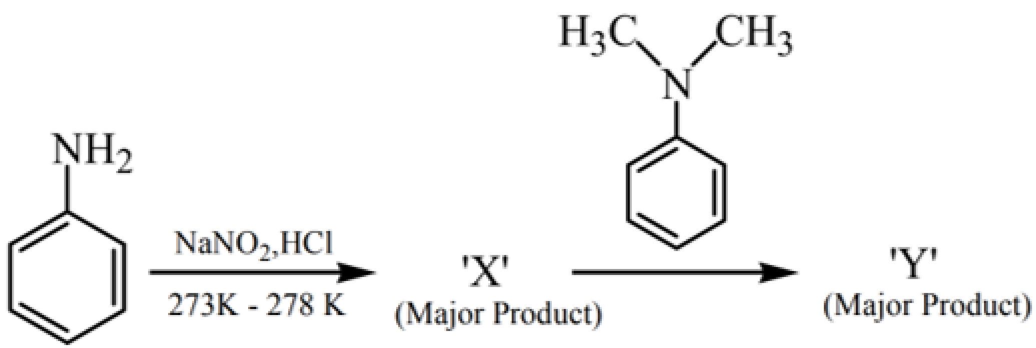


(C)



(D) 

Q43:



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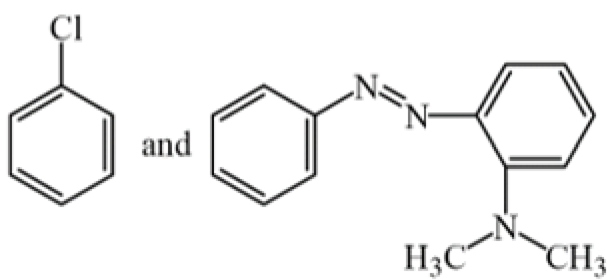
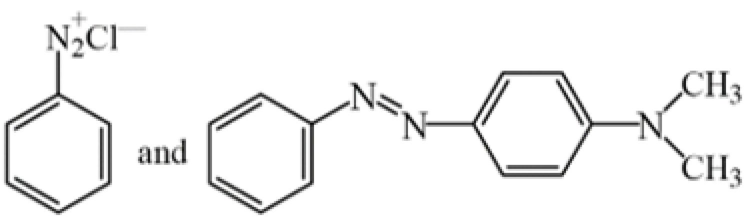
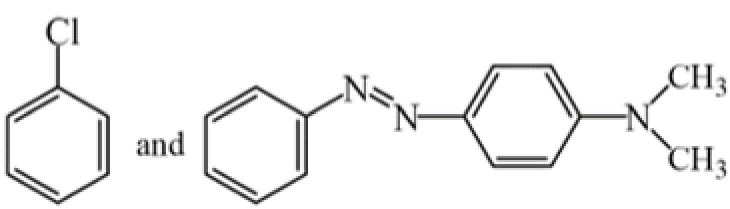


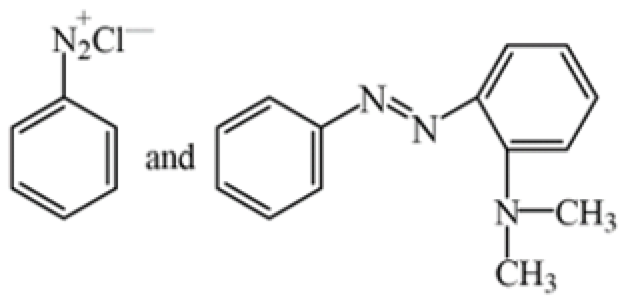
(B)



(C)

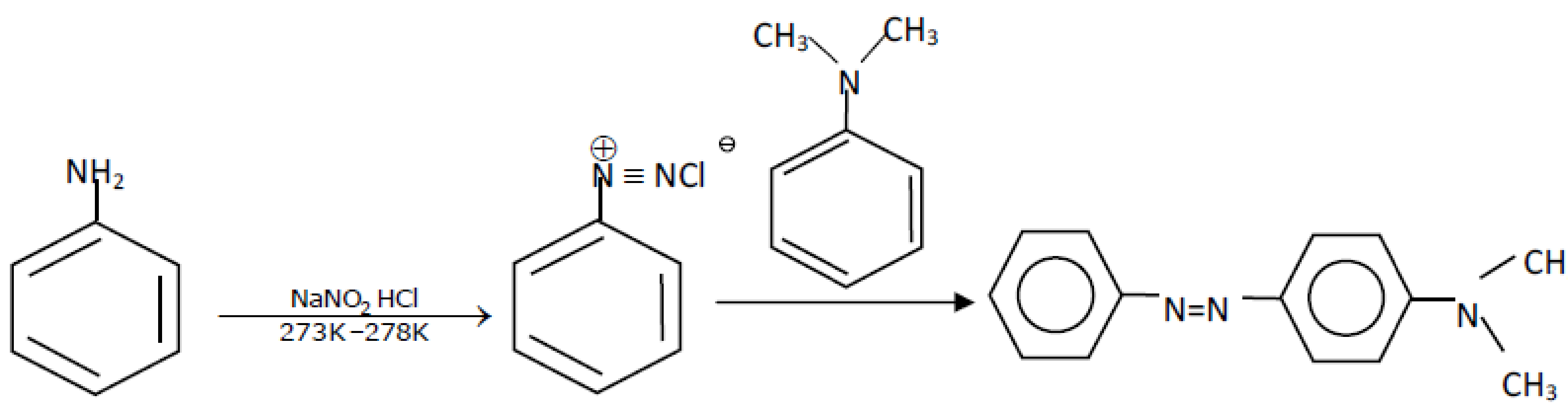


(D)



$\overset{+}{N_2}Cl^-$
and
N
N
N
CH3
CH3

Solution:



NH2
NaNO2 HCl
273K−278K
N ≡ NCl
CH3
CH3
N
N=N
N
CH3
CH3

Q44: The statements that are TRUE:

(A) methane leads to both global warming and photochemical smog

(B) methane is generated from paddy fields

(C) methane is stronger global warming gas than CO_2

(D) methane is a part of reducing somg.

Choose the most appropriate answer from the options given below

(A) (A), (B), (C) only

(B) (A) and (B) only

(C) (A), (B), (D) only

(D) (B), (C), (D) only

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(A) methane leads to both global warming and photochemical smog

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Choose the most appropriate answer from the options given below

(A) (A), (B), (C) only

(B) (A) and (B) only

(C) (A), (B), (D) only

(D) (B), (C), (D) only

Solution:

Contribution of global warming gas

$CO_2 > CH_4 > CFC > O_3 > N_2O > H_2O$

But CH_4 is 40 times stronger green house gases than CO_2 it has more heating effect.

Q45: Reagent, 1-naphthylamine and sulphanilic acid in acetic acid is used for the detection of :

(A) NO_2^-

(B) N_2O

(C) NO_3^-

(D) NO

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(A) NO_2^-

(B) N_2O

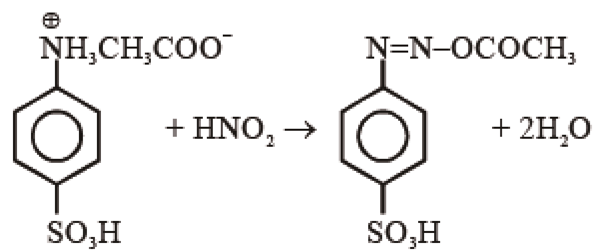
(C) NO_3^-

(D) NO

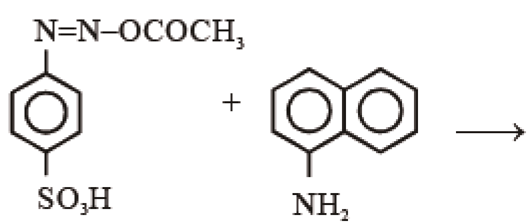
Solution:

For detection of NO^-_2, the following test is used.

$NO^-_2 + CH_3COOH \rightarrow HNO_2 + CH_3COO^-$

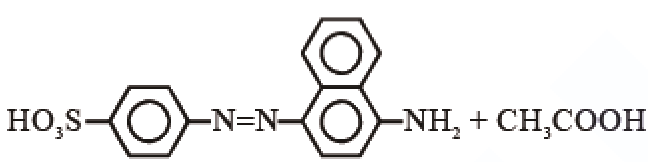


(Sulphanilic acid solution)

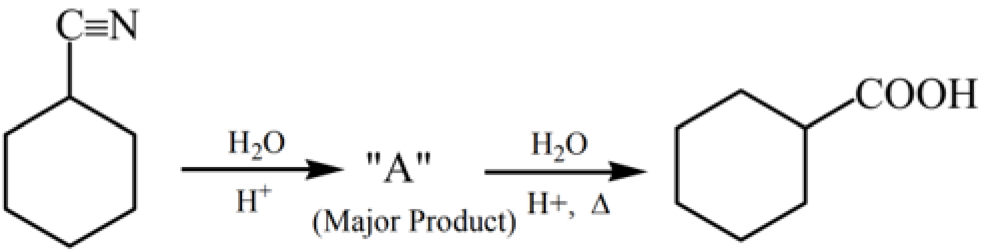


Diazotized acid

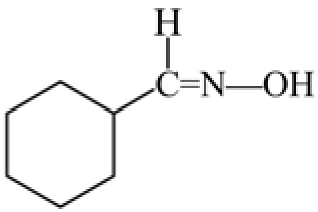
1-napthyl amine

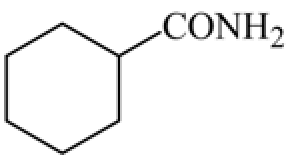


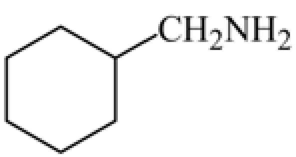
(Red azo dye)

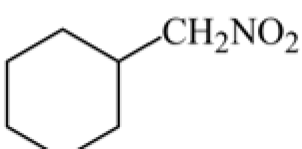
Q46: 

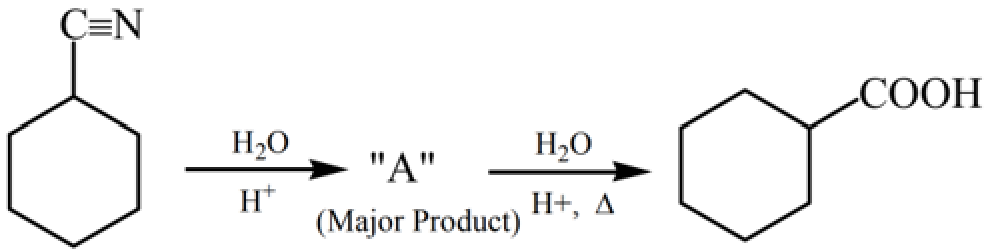
Consider the above chemical reaction and identify product “A” :

(A) 

(B) 

(C) 

(D) 

Q46: 

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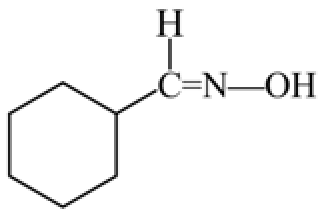
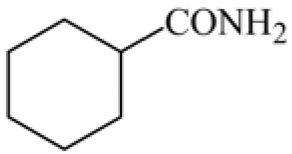
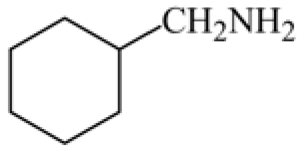
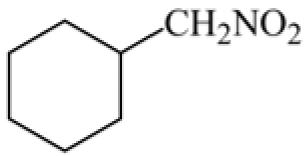
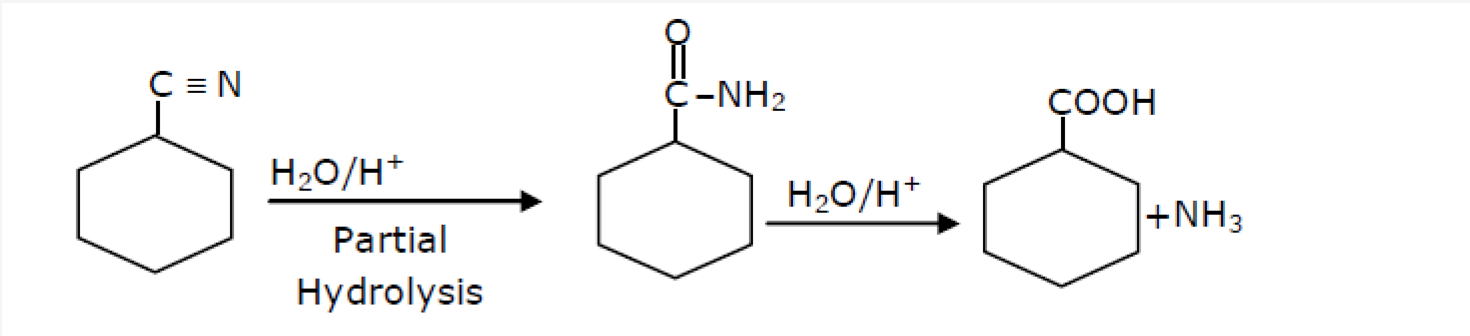
(A) 

(B) 

(C) 

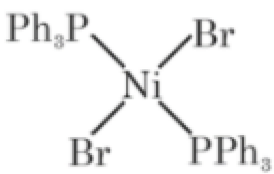
(D) 

Solution:

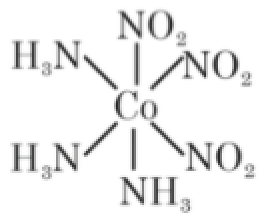


Q47: The correct structures of trans-[$NiBr_2(PPH_3)_2$] and meridonial-[$Co(NH_3)_3(NO_2)_3$], respectively, are :

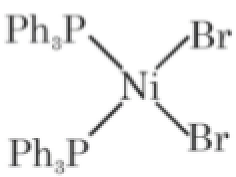
(A)



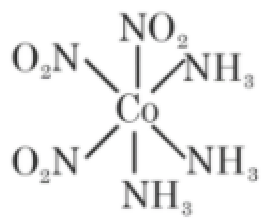
and



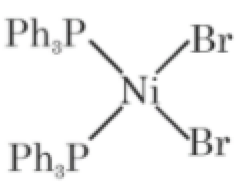
(B)



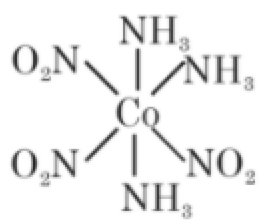
and



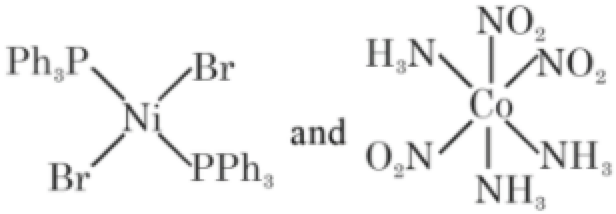
(C)



and



(D)



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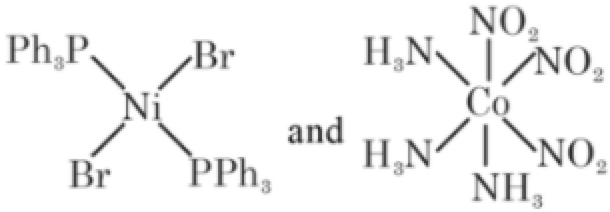
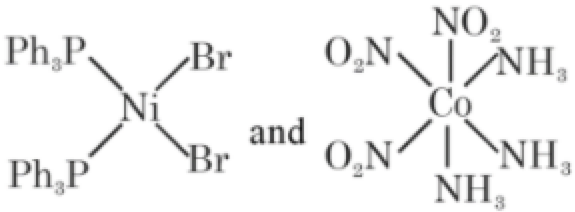
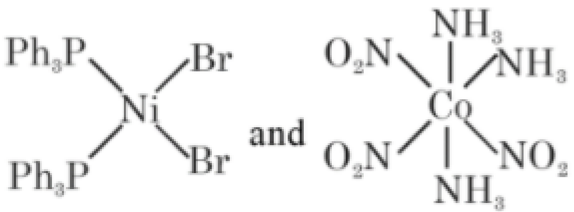
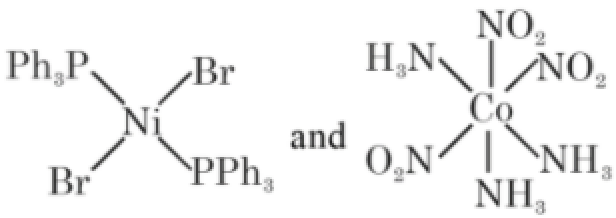


(B)



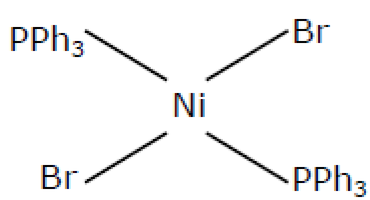
(C)



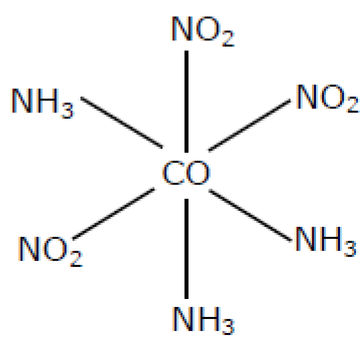
(D) 

Solution:

Trans [Ni $Br_2(Pph_3)_2$]



Meridonial [$CO(NH_3)_3(NO_2)_3$]



Q48: Match List-I with List-II

List-I	**List-II**
(A) Chlorophyll	**(i)** Ruthenium
(B) Vitamin – B_{12}	**(ii)** Platinum
(C) Anticancer drug	**(iii)** Cobalt
(D) Grubbs catalyst	**(iv)** Magnesium

Choose the most appropriate answer from the options given below:

(A) (A) – (iii), (B) – (ii), (C) – (iv), (D) – (i)

(B) (A) – (iv), (B) – (iii), (C) – (ii), (D) – (i)

(C) (A) – (iv), (B) – (ii), (C) – (iii), (D) – (i)

(D) (A) – (iv), (B) – (iii), (C) – (i), (D) – (ii)

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(B) (A) – (iv), (B) – (iii), (C) – (ii), (D) – (i)

(C) (A) – (iv), (B) – (ii), (C) – (iii), (D) – (i)

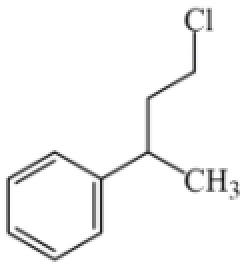
(D) (A) – (iv), (B) – (iii), (C) – (i), (D) – (ii)

Solution:

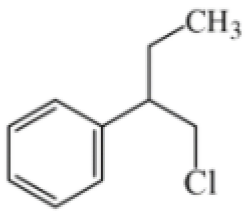
Cis - Platin $[Pt(NH_3)_2Cl_2]$ used in treatment of cancer.
Chlorophyll is complex of Mg
Vitamin B_{12} is a complex of Co
Grubb's catalyst are a series of catalyst containing ruthenium

Q49: Reaction of Grignard reagent, C_2H_5MgBr with C_8H_8O followed by hydrolysis gives compound “A” which reacts instantly with Lucas reagent to give compound B, $C_{10}H_{13}Cl$. The Compound B is:

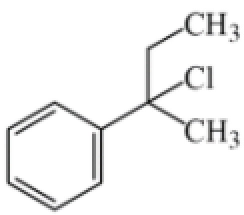
(A)



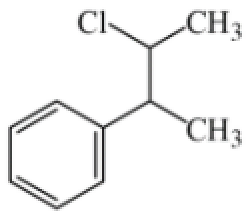
(B)



(C)



(D)



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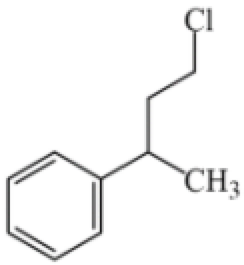
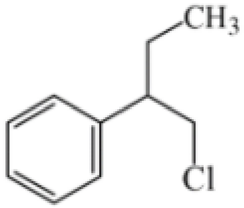
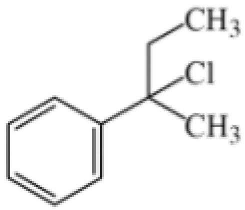
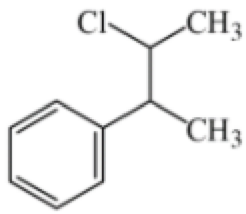


(B)



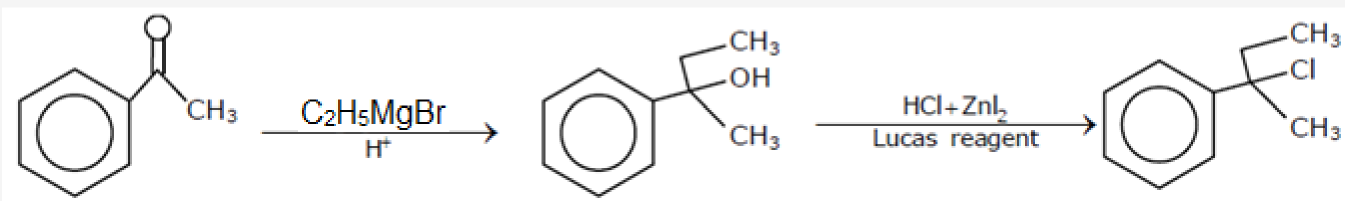
(C)



(D) 

Solution:

$$C_8H_8O \xrightarrow[H^+]{C_2H_5MgBr} (A) \xrightarrow{\text{Lucas reagent}} (B)\ C_{10}H_{13}Cl$$



Q50: In a binary compound, atoms of element A form a hcp structure and those of element M occupy 2/3 of the tetrahedral voids of the hcp structure. The formula of the binary compound is:

(A) M_2A_3

(B) M_4A_3

(C) M_4A

(D) MA_3

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(A) M_2A_3

(B) M_4A_3

(C) M_4A

(D) MA_3

Solution:

A → hcp

M → $2/3^{rd}$ of tetrahedral

$M_{2\times12/3}\,A_6 = M_8A_6 = M_4A_3$

Q51: For the reaction

$$2Fe^{3+}\ (aq)\ +\ 2I^{-}\ (aq)\ \rightarrow\ 2Fe^{2+}\ (aq)\ +\ I_2\,(s)$$

the magnitude of the standard molar free energy change,

$\Delta_r G^o_m\ = -\ \ldots\ldots\ldots\ldots\ldots\ldots\ kJ$ (Round off to the Nearest Integer).

$$\begin{bmatrix} E^o_{Fe^{2+}/Fe\,(s)} = -\,0.440\ V; & E^o_{Fe^{3+}/Fe\,(s)} = -\,0.036\,V \\ E^o_{I_2/2I^-} = 0.539\ V; & F = 96500\ C \end{bmatrix}$$

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45.00

Solution:

$2Fe^{3+}$ (aq) + $2I^-$(aq) $\rightarrow$ $2Fe^{+2}$(aq) + I_2(s)

$1 \times E^o_{Fe^{+3}/Fe^{+2}} + 2 \times E^o_{Fe^{+2}/Fe} = 3 \times E^o_{Fe^{+3}/Fe}$

$E^o_{Fe^{+3}/Fe^{+2}} = 3 \times (-0.036) - 2 \times (-0.44)$

$= -0.108 + 0.88 = 0.772$ V

$E^o_{cell} = E^o_{Fe^{+3}/Fe^{+2}} + E^o_{I^-/I_2}$

$= 0.772 - 0.539 = 0.233$ V

$\Delta G^o = nFE^o_{cell}$

$= +2 \times 96500 \times 0.233$

$= 44969$ J $= 44.9$ kJ $= 45$ kJ

Q52: Complete combustion of 3g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is .... (Round off to the Nearest Integer).

[Use: $N_A = 6.023 \times 10^{23}$; Atomic masses in u : C : 12.0 ; O : 16.0 ; H : 1.0]

Q52: Complete combustion of 3g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is .... (Round off to the Nearest Integer).
[Use: $N_A = 6.023 \times 10^{23}$; Atomic masses in u : C : 12.0 ; O : 16.0 ; H : 1.0]

18.00

Solution:

$C_2H_6 + O_2 \rightarrow 2CO_2 + 3H_2O$
3gm 0.3 mol
0.1 mol 0.3 N_A
$= 0.3 \times 6.023 \times 10^{23}$ molecules of H_2O
$= 1.8069 \times 10^{23}$
$= 18.069 \times 10^{22}$

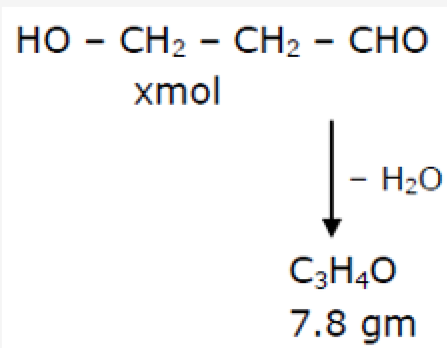
Q53: .... grams of 3-hydroxy propanal (MW = 74) must be dehydrated to produce 7.8 g of acrolein (MW = 56) (C_3H_4O) if the percentage yield is 64. (Round off to the Nearest Integer).
[Given: Atomic masses: C : 12.0 u, H : 1.0 u, O : 16.0 u]

Q53: .... grams of 3-hydroxy propanal (MW = 74) must be dehydrated to produce 7.8 g of acrolein (MW = 56) (C_3H_4O) if the percentage yield is 64. (Round off to the Nearest Integer).

[Given: Atomic masses: C : 12.0 u, H : 1.0 u, O : 16.0 u]

16.00

Solution:



$\frac{7.8}{50}$ = 0.14 mol

% yield = $\frac{7.8/56}{x}$ × 100 = 64

$$x = \frac{7.8 \times 100}{56 \times 64} = \frac{780}{56 \times 64} \text{mol}$$

$W_{Reactant}$ = $\frac{780}{56 \times 64}$ × 74 = 16.11 gm

Q54: $2\ NO\,(g) + \ Cl_2\,(g) \ = \ 2\ NOCl\,(s)$

This reaction was studied at – 10ºC and the following data was obtained

run	**$[NO]_0$**	**$[Cl_2]_0$**	**r_0**
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate.

The overall of the reaction is .... (Round off to the Nearest Integer).

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1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate.

The overall of the reaction is .... (Round off to the Nearest Integer).

3.00

Solution:

Exp. (I) 0.18 = K(0.1)^X(0.I)$^{+Y}$....(1)

Exp. (II) 0.35 = K(0.1)^X(0.2)^Y....(2)

Exp. (III) 1.40 = K(0.2)^X(0.2)^Y....(3)

(2) ÷ (3)

$\frac{0.35}{1.40} = \frac{\mathrm{K} \times (0.1)^{\mathrm{x}} (0.2)^{\mathrm{y}}}{\mathrm{K} (0.2)^{\mathrm{x}} (0.2)^{\mathrm{y}}}$

$\frac{1}{4} = \left(\frac{1}{2}\right)^{\mathrm{x}} \Rightarrow \mathrm{X} = 2$

(1) ÷ (1)

$\frac{1}{2} = \left(\frac{1}{2}\right)^{\mathrm{y}} \Rightarrow \mathrm{Y} = 1$

Q55: 2 molal solution of a weak acid HA has a freezing point of 3.885ºC. The degree of dissociation of this acid is .... $\times 10^{-3}$. (Round off to the Nearest Integer).

[Given: Molal depression constant of water = 1.85 K kg mol^{-1}

Freezing point of pure water = 0ºC]

Q55: 2 molal solution of a weak acid HA has a freezing point of 3.885ºC. The degree of dissociation of this acid is .... $\times 10^{-3}$. (Round off to the Nearest Integer).
[Given: Molal depression constant of water = 1.85 K kg mol^{-1}
Freezing point of pure water = 0ºC]

50.00

Solution:

$T_{f\ so1.} = -3.885°C$
$\Delta T_f = +3.885 = i \times k_f \times m$
$3.885 = i \times 1.85 \times 2$
$i = \frac{3.885}{1.85 \times 2} = [1 + \propto]$
$\propto = \frac{0.185}{3.7} = 0.05 = 50 \times 10^{-3}$
Answer is 50

Q56: AX is a covalent diatomic molecule where A and X are second row elements of periodic table. Based on Molecular orbital theory, the bond order of AX is 2.5. The total number of electrons in AX is .... (Round off to the Nearest Integer).

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15.00

Solution:

The comp. AX is NO its bond order is 2.5 & it has total 15 electrons

Q57: In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is .... M. (Round off to the Nearest Integer).
[Given: pKa (acetic acid) = 4.74]

Q57: In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is .... M. (Round off to the Nearest Integer).
[Given: pKa (acetic acid) = 4.74]

10.00

Solution:

Buffer pH = 5.74

$$= \mathrm{p}^{\mathrm{Ka}}_{\text{acetic acid}} + \log\left[\frac{\text{Sodium acetate}}{\text{Acetic acid}}\right]$$

$$\frac{\text{Sodium acetate}}{\text{Acetic acid}} = 10$$

Sodium acetate = 10 M

Q58: The total number of unpaired electrons present in the complex $K_3[Cr(oxalate)_3]$ is

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3.00

Solution:

$K_3[Cr(C_2O_4)_3]$
Chromium is in + 3 oxidation state
$Cr \rightarrow 3d^5\ 4s^1$
$Cr^{3+} \rightarrow 3d^3$ = Three unpaired electrons
The hybridisation of chromium in the complex is d^2sp^3

Q59: A reaction of 0.1 mole of Benzylamine with bromomethane gave 23 g of Benzyl trimethyl ammonium bromide. The number of moles of bromomethane consumed in this reaction are n $\times 10^{-1}$, when n = ... (Round off to the Nearest Integer).

[Given: Atomic masses: C : 12.0 u, H : 1.0 u, N : 14.0 u, Br : 80.0 u]

Q59: A reaction of 0.1 mole of Benzylamine with bromomethane gave 23 g of Benzyl trimethyl ammonium bromide. The number of moles of bromomethane consumed in this reaction are n $\times 10^{-1}$, when n = ... (Round off to the Nearest Integer).
[Given: Atomic masses: C : 12.0 u, H : 1.0 u, N : 14.0 u, Br : 80.0 u]

3.00

Solution:

$Ph - CH_2NH_2 + 3CH_2Br \rightarrow PhCH_2N^+(Me)_3Br^-$

0.1 mol $\frac{23\ g}{230}$ = 0.1 mol

$\therefore$ moles of CH_3Br = 0.3 = 3×10^{-1} mol

Q60: For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$
the reaction enthalpy $\Delta_r H = \ldots\ldots\ldots\ldots\ kJ\ mol^{-1}$. (Round off to the Nearest Integer).
[Given: Bond enthalpies in kJ mol^{-1} : C – C : 347, C = C : 611; C – H : 414, H – H : 436]

Q60: For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$
the reaction enthalpy $\Delta_r H = \ldots\ldots\ldots\ldots \; kJ\ mol^{-1}$. (Round off to the Nearest Integer).
[Given: Bond enthalpies in kJ mol^{-1} : C – C : 347, C = C : 611; C – H : 414, H – H : 436]

128.00

Solution:

$C_2H_6 \rightarrow C_2H_4 + H_2$
$\Delta H = ?$
$= 2 \times E_{C-H} - E_{C=H} - E_{H-H} + E_{C-C}$
$= 2 \times 414 + 347 - 611 - 436$
$= 828 + 347 - 1047$
$= 128$ kJ/mol

Q61: If the equation $a\,|z|^2 + \overline{\bar{\alpha} z + \alpha \bar{z}} + d = 0$ represents a circle where a, d are real constants, then which of the following condition is correct ?

(A) $|\alpha|^2 - ad \neq 0$

(B) $|\alpha|^2 - ad \geqslant 0\ and\ a \in R$

(C) $\alpha = 0,\ a,\ d \in R^+$

(D) $|\alpha|^2 - ad > 0$ and $a \in R - \{0\}$

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(D) $|\alpha|^2 - ad > 0 \text{ and } a \in R - \{0\}$

Solution:

$az\bar{z} + \alpha\bar{z} + \bar{\alpha}z + d = 0 \quad \rightarrow$ Circle

Centre $= \frac{-\alpha}{a}\ 2 = \sqrt{\frac{\alpha\bar{\alpha}}{a^2} - \frac{d}{a}} = \sqrt{\frac{\alpha\bar{\alpha} - ad}{a^2}}$

So, $|\alpha|^2 - ad > 0$ and $a \in R - \{0\}$

Q62: The solutions of the equation

$$\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0, \ (0 < x < \pi),$$ are :

(A) $\frac{7\pi}{12}, \frac{11\pi}{12}$

(B) $\frac{\pi}{12}, \frac{\pi}{6}$

(C) $\frac{5\pi}{12}, \frac{7\pi}{12}$

(D) $\frac{\pi}{6}, \frac{5\pi}{6}$

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(B) $\frac{\pi}{12}, \frac{\pi}{6}$

(C) $\frac{5\pi}{12}, \frac{7\pi}{12}$

(D) $\frac{\pi}{6}, \frac{5\pi}{6}$

Solution:

$$\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\ \sin\ 2x & 4\ \sin\ 2x & 1+4\ \sin\ 2x \end{vmatrix} = \ 0$$

Use $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow\ (2+4\sin 2x)\begin{vmatrix} 1 & 1 & 1 \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0$$

$\Rightarrow\ \sin 2x = -\frac{1}{2}$

$\Rightarrow\ 2x = \pi + \frac{\pi}{6},\ 2\pi - \frac{\pi}{6}$

$x = \frac{\pi}{2} + \frac{\pi}{12},\ \pi - \frac{\pi}{12}$

Q63: If $f(x) = \begin{cases} \frac{1}{|x|} & ; \quad |x| \geqslant 1 \\ ax^2 + b & ; \quad |x| < 1 \end{cases}$ is differentiable at every point of the domain, then the value of a and b are respectively :

(A) $\frac{5}{2}, -\frac{3}{2}$

(B) $\frac{1}{2}, -\frac{3}{2}$

(C) $-\frac{1}{2}, \frac{3}{2}$

(D) $\frac{1}{2}, \frac{1}{2}$

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(D) $\frac{1}{2}, \frac{1}{2}$

Solution:

$$f(x) = \begin{cases} \frac{1}{|x|}, & |x| \geqslant 1 \\ ax^2 + b, & |x| < 1 \end{cases}$$

At $x = 1$ function must be continuous

So, $1 = a + b \qquad \ldots .(1)$

Differentiability at $x = 1$

$$\left(-\frac{1}{x^2}\right)_{x=1} = (2ax)_{x=1}$$

$\Rightarrow \ -1 = 2a \ \Rightarrow \ a = -\frac{1}{2}$

From $(1) \ \Rightarrow \ b = 1 + \frac{1}{2} = \frac{3}{2}$

Q64: The number of integral values of m so that the abscissa of point of intersection of lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer, is :

(A) 1

(B) 3

(C) 0

(D) 2

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(A) 1

(B) 3

(C) 0

(D) 2

Solution:

$3x + 4y = 9$
$y = mx + 1$
$\Rightarrow \ 3x + 4mx + 4 = 9$
$\Rightarrow \ (3 + 4m)\,x = 5$
$\Rightarrow$ x will be an integer when
$3 + 4m = 5, \ -5, \ 1, \ -1$
$\Rightarrow \ m = \frac{1}{2}, \ -2, \ -\frac{1}{2}, \ -1$
So, number of integral values of m is 2

Q65: Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$. If $T_r(A)$ denotes the sum of all diagonal elements of the matrix A, then $T_r(A) - T_r(B)$ has value equal to :

(A) 3

(B) 1

(C) 2

(D) 0

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(A) 3

(B) 1

(C) 2

(D) 0

Solution:

$$A+2B=\begin{pmatrix}1 & 2 & 0\\ 6 & -3 & 3\\ -5 & 3 & 1\end{pmatrix} \qquad \ldots(1)$$

$$2A-B=\begin{pmatrix}2 & -1 & 5\\ 2 & -1 & 6\\ 0 & 1 & 2\end{pmatrix}$$

$$\Rightarrow\ 4A-2B=\begin{pmatrix}4 & -2 & 10\\ 4 & -2 & 12\\ 0 & 2 & 4\end{pmatrix} \qquad \ldots.(2)$$

$$(1)+(2)\ \Rightarrow\ 5A=\begin{pmatrix}5 & 0 & 10\\ 10 & -5 & 15\\ -5 & 5 & 5\end{pmatrix}$$

$$A=\begin{pmatrix}1 & 0 & 2\\ 2 & -1 & 3\\ -1 & 1 & 1\end{pmatrix} \text{ and } 2A=\begin{pmatrix}2 & 0 & 4\\ 4 & -2 & 6\\ -2 & 2 & 2\end{pmatrix}$$

$$\therefore\ B=\begin{pmatrix}2 & 0 & 4\\ 4 & -2 & 6\\ -2 & 2 & 2\end{pmatrix}-\begin{pmatrix}2 & -1 & 5\\ 2 & -1 & 6\\ 0 & 1 & 2\end{pmatrix}$$

$$B=\begin{pmatrix}0 & 1 & -1\\ 2 & -1 & 0\\ -2 & 1 & 0\end{pmatrix}$$

$tr\,(A)=1-1+1=1$

$tr\,(B)=-1$

$tr\,(A)=1$ and $tr\,(B)=-1$

$\therefore\ tr\,(A)-tr\,(B)=2$

Q66: The integral $\int \frac{(2x-1)\ \cos\ \sqrt{(2x-1)^2\ +\ 5}}{\sqrt{4x^2-4x\ +\ 6}}\ dx$ is equal to :

(where c is a constant of integration)

(A) $\frac{1}{2}\cos\ \sqrt{(2x-1)^2\ +\ 5}\ +\ c$

(B) $\frac{1}{2}\sin\ \sqrt{(2x+1)^2\ +\ 5}\ +\ c$

(C) $\frac{1}{2}\cos\ \sqrt{(2x+1)^2\ +\ 5}\ +\ c$

(D) $\frac{1}{2}\sin\sqrt{(2x-1)^2\ +\ 5}\ +\ c$

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(B) $\frac{1}{2}\sin\ \sqrt{(2x+1)^2\ +\ 5}\ +\ c$

(C) $\frac{1}{2}\cos\ \sqrt{(2x+1)^2\ +\ 5}\ +\ c$

(D) $\frac{1}{2}\sin\sqrt{(2x-1)^2\ +\ 5}\ +\ c$

Solution:

$$\int \frac{(2x-1)\cos\sqrt{(2x-1)^2+5}}{\sqrt{(2x-1)^2+5}}dx$$

$$(2x-1)^2+5=t^2$$

$$2(2x-1)\,2dx=2t\,dt$$

$$2\sqrt{t^2-5}\,dx=t\,dt$$

So, $\int \frac{\sqrt{t^2-5}\cos t}{2\sqrt{t^2-5}}dt=\frac{1}{2}\sin t+c$

$$=\frac{1}{2}\sin\sqrt{(2x-1)^2+5}+c$$

Q67: If the functions are defined as $f\ (x) = \ \sqrt{x}$ and $g\ (x) = \ \sqrt{1-x}$, then what is the common domain of the following functions : $f+g,\ \ f-g,\ \ \frac{f}{g},\ \ \frac{g}{f},\ \ g-f$ where $(f \pm\ g)\ (x)\ =\ f\,(x)\ \pm\ g\,(x),\ (f/g)\ (x) = \ \frac{f\,(x)}{g\,(x)}$

(A) $0 \leqslant x < 1$

(B) $0 < x \leqslant 1$

(C) $0 \leqslant x \leqslant 1$

(D) $0 < x < 1$

Q67: If the functions are defined as $f\ (x) = \sqrt{x}$ and $g\ (x) = \sqrt{1-x}$, then what is the common domain of the following functions : $f+g,\ f-g,\ \frac{f}{g},\ \frac{g}{f},\ g-f$ where $(f \pm g)\ (x) = f\ (x) \pm g\ (x),\ (f/g)\ (x) = \frac{f\ (x)}{g\ (x)}$

(A) $0 \leqslant x < 1$

(B) $0 < x \leqslant 1$

(C) $0 \leqslant x \leqslant 1$

(D) $0 < x < 1$

Solution:

$f\ (x) + g\ (x) = \sqrt{x} + \sqrt{1-x}$, domain $[0,\ 1]$
$f\ (x) - g\ (x) = \sqrt{x} - \sqrt{1-x}$, domain $[0,\ 1]$
$g\ (x) - f\ (x) = \sqrt{1-x} - \sqrt{x}$, domain $[0,\ 1]$
$\frac{f(x)}{g(x)} = \frac{\sqrt{x}}{\sqrt{1-x}}$, domain $[0,\ 1)$
$\frac{g(x)}{f(x)} = \frac{\sqrt{1-x}}{\sqrt{x}}$, domain $(0,\ 1]$
So, common domain is $(0,\ 1)$

Q68: The sum of all the 4-digit distinct numbers that can be formed with the digits 1, 2, 2 and 3 is :

(A) 122234

(B) 22264

(C) 122664

(D) 26664

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(A) 122234

(B) 22264

(C) 122664

(D) 26664

Solution:

Digits are 1, 2, 2, 3

Total distinct numbers = $\frac{4!}{2!} = 12$

Total numbers when 1 at unit place is 3

2 at unit place is 6

3 at unit place is 3.

So, sum $= (3 + 12 + 9)\left(10^3 + 10^2 + 10 + 1\right)$

$= (1111) \times 24$

$= 26664$

Q69: Let $(1+x+2x^2)^{20} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{40} x^{40}$. Then, $a_1 + a_3 + a_5 + \ldots\ldots + a_{37}$ is equal to :

(A) $2^{20}\left(2^{20} - 21\right)$

(B) $2^{19}\left(2^{20} + 21\right)$

(C) $2^{19}\left(2^{20} - 21\right)$

(D) $2^{20}\left(2^{20} + 21\right)$

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(B) $2^{19}(2^{20} + 21)$

(C) $2^{19}(2^{20} - 21)$

(D) $2^{20}(2^{20} + 21)$

Solution:

$(1+x+2x^2)^{20} = a_0 + a_1 x + \ldots + a_{40} x^{40}$

Put $x = 1, \ -1$

$\Rightarrow \ a_0 + a_1 + a_2 + \ldots + a_{40} = 4^{20}$

$a_0 - a_1 + a_2 + \ldots + a_{40} = 2^{20}$

$\Rightarrow \ a_1 + a_3 + \ldots + a_{39} = \frac{4^{20} - 2^{20}}{2}$

$\Rightarrow \ a_1 + a_3 + \ldots + a_{37} = 2^{39} - 2^{19} - a_{39}$

Here $a_{39} = \frac{20!(2)^{19} \times 1}{19!} = 20 \times 2^{19}$

$\Rightarrow \ a_1 + a_3 + \ldots + a_{37} = 2^{19}(2^{20} - 1 - 20)$

$= 2^{19}(2^{20} - 21)$

Q70: If $\lim_{x \to 0} \frac{\sin^{-1} x - \tan^{-1} x}{3x^3}$ is equal to L, then the value of $(6L+1)$ is :

(A) $\frac{1}{6}$

(B) 6

(C) 2

(D) $\frac{1}{2}$

Q70: If $\lim\limits_{x\to\ 0} \frac{\sin^{-1}x-\ \tan^{-1}x}{3x^3}$ is equal to L, then the value of $(6L+1)$ is :

(A) $\frac{1}{6}$

(B) 6

(C) 2

(D) $\frac{1}{2}$

Solution:

$$L=\lim_{x\to 0}\frac{\left(x+\frac{x^3}{3!}\ldots\right)-\left(x-\frac{x^3}{3}\ldots\right)}{3x^3}=\frac{1}{6}$$

So $6L+1=2$

Q71: The equation of one of the straight lines which passes through the point (1, 3) and makes and angle $\tan^{-1}(\sqrt{2})$ with the straight line, $y + 1 = 3\sqrt{2}x$ is :

(A) $4\sqrt{2}x + 5y - 4\sqrt{2} = 0$

(B) $4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$

(C) $5\sqrt{2}x + 4y - (15 + 4\sqrt{2}) = 0$

(D) $4\sqrt{2}x - 5y - (5 + 4\sqrt{2}) = 0$

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(A) $4\sqrt{2}x + 5y - 4\sqrt{2} = 0$

(B) $4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$

(C) $5\sqrt{2}x + 4y - (15 + 4\sqrt{2}) = 0$

(D) $4\sqrt{2}x - 5y - (5 + 4\sqrt{2}) = 0$

Solution:

$y = mx + c$

$3 = m + c$

$\sqrt{2} = \left|\frac{m-3\sqrt{2}}{1+3\sqrt{2}m}\right|$

$= 6m + \sqrt{2} = m - 3\sqrt{2} \rightarrow m = \frac{-4\sqrt{2}}{5}$

$= -6m - \sqrt{2} = m - 3\sqrt{2} \rightarrow m = \frac{2\sqrt{2}}{7}$

According to options take $m = \frac{-4\sqrt{2}}{5}$

So $y = \frac{-4\sqrt{2}x}{5} + \frac{15+4\sqrt{2}}{5}$

$4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$

Q72: A vector $\vec{a}$ has components 3p and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components $p+1$ and $\sqrt{10}$, then a value of p is equal to :

(A) 1

(B) $\frac{4}{5}$

(C) $-\frac{5}{4}$

(D) –1

Q72: A vector $\vec{a}$ has components 3p and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components $p+1$ and $\sqrt{10}$, then a value of p is equal to :

(A) 1

(B) $\frac{4}{5}$

(C) $-\frac{5}{4}$

(D) –1

Solution:

$$\bar{a}_{old} = 3p\hat{i} + \hat{j}$$
$$\bar{a}_{new} = (p+1)\,\hat{i} + \sqrt{10}\hat{j}$$
$$\Rightarrow\ |\bar{a}_{old}| = |\bar{a}_{new}|$$
$$8p^2 - 2p - 10 = 0$$
$$4p^2 - p - 5 = 0$$
$$(4p-5)\,(p+1) = 0 \ \rightarrow\ p = \tfrac{5}{4},\ -1$$

Q73: Choose the correct statement about two circles whose equations are given below :

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$x^2 + y^2 - 22x - 10y + 137 = 0$$

(A) circles have same centre

(B) circles have no meeting point

(C) circles have only one meeting point

(D) circles have two meeting points

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$$x^2 + y^2 - 10x - 10y + 41 = 0$$
$$x^2 + y^2 - 22x - 10y + 137 = 0$$

(A) circles have same centre

(B) circles have no meeting point

(C) circles have only one meeting point

(D) circles have two meeting points

Solution:

$x^2 + y^2 - 10x - 10y + 41 = 0$
$A(5,\ 5),\ R_1 = 3$
$x^2 + y^2 - 22x - 10y + 137 = 0$
$B(11,\ 5),\ R_2 = 3$
$AB = 6 = R_1 + R_2$
Touch each other externally
$\Rightarrow$ circles have only one meeting point

Q74: For the four circles M, N, O and P, following four equations are given :

Circle M : $x^2 + y^2 = 1$

Circle N : $x^2 + y^2 - 2x = 0$

Circle O : $x^2 + y^2 - 2x - 2y + 1 = 0$

Circle P : $x^2 + y^2 - 2y = 0$

If the centre of circle M is joined with centre of the circle N, further centre of circle N is joined with centre of the circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, then these lines form the sides of a :

(A) Rhombus

(B) Rectangle

(C) Parallelogram

(D) Square

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Circle M : $x^2 + \ y^2 = 1$

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Circle O : $x^2 + \ y^2 - 2x - 2y + 1 = \ 0$

Circle P : $x^2 + y^2 - 2y = \ 0$

If the centre of circle M is joined with centre of the circle N, further centre of circle N is joined with centre of the circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, then these lines form the sides of a :

(A) Rhombus

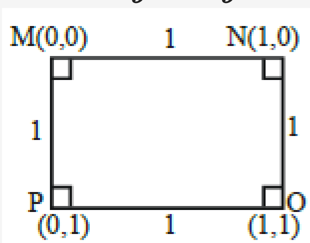
(B) Rectangle

(C) Parallelogram

(D) Square

Solution:

$$M : x^2 + y^2 = 1 \qquad\qquad (0,\ 0)$$
$$N : x^2 + y^2 - 2x = 0 \qquad\qquad (1,\ 0)$$
$$O : x^2 + y^2 - 2x - 2y + 1 = 0 \qquad\qquad (1,\ 1)$$
$$P : x^2 + y^2 - 2y = 0 \qquad\qquad (0,\ 1)$$



Q75: The value of $3 + \cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+\ldots\ldots\ldots\ \infty}}}}$ is equal to :

(A) $2 + \sqrt{3}$

(B) $4 + \sqrt{3}$

(C) $3 + 2\sqrt{3}$

(D) $1.5 + \sqrt{3}$

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(A) $2 + \sqrt{3}$

(B) $4 + \sqrt{3}$

(C) $3 + 2\sqrt{3}$

(D) $1.5 + \sqrt{3}$

Solution:

Let $x = 3 + \cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+\ldots\ldots\ldots\infty}}}}$

So, $x = 3 + \frac{1}{4+\frac{1}{x}} = 3 + \frac{1}{\frac{4x+1}{x}}$

$\Rightarrow \ (x-3) = \frac{x}{(4x+1)}$

$\Rightarrow \ (4x+1)(x-3) = x$

$\Rightarrow \ 4x^2 - 12x + x - 3 = x$

$\Rightarrow \ 4x^2 - 12x - 3 = 0$

$x = \frac{12 \pm \sqrt{(12)^2 + 12 \times 4}}{2 \times 4} = \frac{12 \pm \sqrt{12(16)}}{8}$

$= \frac{12 \pm 4 \times 2\sqrt{3}}{8} = \frac{3 \pm 2\sqrt{3}}{2}$

$x = \frac{3}{2} \pm \sqrt{3} = 1.5 \pm \sqrt{3}$

But only positive value is accepted

So, $x = 1.5 + \sqrt{3}$

Q76: The differential equation satisfied by the system of parabolas $y^2 = 4a\,(x + a)$ is :

(A) $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

(B) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) + y = 0$

(C) $y\left(\frac{dy}{dx}\right) + 2x\left(\frac{dy}{dx}\right) - y = 0$

(D) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) - y = 0$

Q76: The differential equation satisfied by the system of parabolas $y^2 = 4a(x+a)$ is :

(A) $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

(B) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) + y = 0$

(C) $y\left(\frac{dy}{dx}\right) + 2x\left(\frac{dy}{dx}\right) - y = 0$

(D) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) - y = 0$

Solution:

$y^2 = 4ax + 4a^2$

Differentiate with respect to x

$\Rightarrow\ 2y\frac{dy}{dx} = 4a$

$\Rightarrow\ a = \left(\frac{y}{2}\frac{dy}{dx}\right)$

So, required differential equation is

$y^2 = \left(4 \times \frac{y}{2}\frac{dy}{dx}\right)x + 4\left(\frac{y}{2}\frac{dy}{dx}\right)^2$

$\Rightarrow\ y^2\left(\frac{dy}{dx}\right)^2 + 2xy\left(\frac{dy}{dx}\right) - y^2 = 0$

$\Rightarrow\ y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

Q77: $\frac{1}{3^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + \ldots\ldots\ldots + \frac{1}{(201)^2-1}$ is equal to :

(A) $\frac{101}{408}$

(B) $\frac{101}{404}$

(C) $\frac{99}{400}$

(D) $\frac{25}{101}$

Q77: $\frac{1}{3^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + \ldots\ldots\ldots + \frac{1}{(201)^2-1}$ is equal to :

(A) $\frac{101}{408}$

(B) $\frac{101}{404}$

(C) $\frac{99}{400}$

(D) $\frac{25}{101}$

Solution:

$$T_n = \frac{1}{(2n+1)^2-1} = \frac{1}{4(n)(n+1)}$$
$$= \frac{(n+1)-n}{4n(n+1)} = \frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right)$$
$$S = \frac{1}{4}\left(1 - \frac{1}{101}\right) = \frac{1}{4}\left(\frac{100}{101}\right) = \frac{25}{101}$$

Q78: Let $\alpha,\ \beta,\ \gamma$ be the real roots of the equation $x^3 + ax^2 + bx + c = 0$,$(a,\ b,\ c\ \in\ R$ and $a,\ b\ \neq\ 0)$. If the system of equation $(\text{in } u,\ v,\ w)$ given by $\alpha u\ +\ \beta v\ +\ \gamma w =\ 0;$ $\beta u\ +\ \gamma v +\ \alpha w = 0; \gamma u +\ \alpha v\ +\ \beta w =\ 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is :

(A) 5

(B) 1

(C) 3

(D) 0

Q78: Let $\alpha,\ \beta,\ \gamma$ be the real roots of the equation $x^3 + ax^2 + bx + c = 0,(a,\ b,\ c\ \in\ R$ and $a,\ b\ \neq\ 0)$. If the system of equation $(\text{in } u,\ v,\ w)$ given by $\alpha u\ +\ \beta v\ +\ \gamma w =\ 0;$ $\beta u\ +\ \gamma v +\ \alpha w = 0; \gamma u +\ \alpha v\ +\ \beta w =\ 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is :

(A) 5

(B) 1

(C) 3

(D) 0

Solution:

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0$$

$$\Rightarrow\ -(\alpha+\beta+\gamma)\left(\alpha^2+\beta^2+\gamma^2-\Sigma\alpha\beta\right)=0$$

$$\Rightarrow\ -(-a)\left(a^2-2b-b\right)=0$$

$$\Rightarrow\ a\left(a^2-3b\right)=0$$

$$\Rightarrow\ a^2=3b$$

$$\Rightarrow\ \frac{a^2}{b}=3$$

Q79: The real valued function $f\ (x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{x - \ [x]}}$, where $[x]$ denotes the greatest integer less than or equal to x, is defined for all x belonging to :

(A) all integers except 0, –1, 1

(B) all reals except the interval [–1, 1]

(C) all reals except integers

(D) all non-integers except the interval [–1, 1]

Q79: The real valued function $f\ (x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{x - \ [x]}}$, where $[x]$ denotes the greatest integer less than or equal to x, is defined for all x belonging to :

(A) all integers except 0, –1, 1

(B) all reals except the interval [–1, 1]

(C) all reals except integers

(D) all non-integers except the interval [–1, 1]

Solution:

$f\ (x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{\{x\}}}$

Domain $\in \left(-\infty,\ -1\right] \cup \left[1,\ \infty\right)$

$\{x\} \neq 0$ so $x \neq$ integers

Q80: If $\alpha,\ \beta$ are natural numbers such that $100^{\alpha} - \ 199 \beta = (100)(100) + (99)(101) + (98)(102) + \ldots + (1)(199)$, then the slope of the line passing through $(\alpha,\ \beta)$ and origin is

(A) 540

(B) 510

(C) 550

(D) 530

Q80: If $\alpha,\ \beta$ are natural numbers such that $100^{\alpha} - \ 199$
$\beta = (100)\,(100) + (99)\,(101) + (98)\,(102) + \ldots + (1)\,(199)$, then the slope of the line passing through $(\alpha,\ \beta)$ and origin is

(A) 540

(B) 510

(C) 550

(D) 530

Solution:

$S = (100)\,(100) + (99)\,(101) + (98)\,(102) + \ldots + (2)\,(198) + (1)\,(199)$

$S = \sum\limits_{x=0}^{99} (100 - x)\,(100 + x) = \sum\limits_{x=0}^{99} 100^2 - x^2$

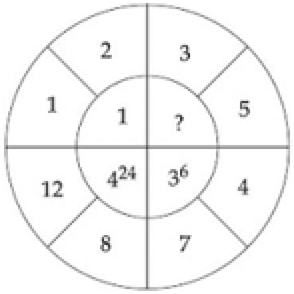
$= 100^3 - \frac{99 \times 100 \times 199}{6}$

$\alpha = 3$

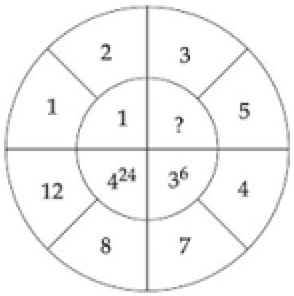
$\beta = 1650$

Slope $= \frac{1650}{3} = 550$

Q81: The missing value in the following figure is ....

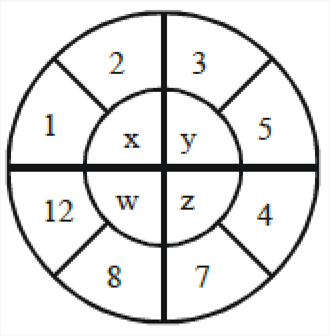


Q81: The missing value in the following figure is ....



4.00

Solution:



$x=(2-1)^{1!}=1$

$w=(12-8)^{4!}=4^{24}$

$z=(7-4)^{3!}=3^{6}$

Hence $y=(5-3)^{2!}=2^{2}$

Q82: Let $f(x)$ and $g(x)$ be two functions satisfying $f\left(x^2\right)+g(4-x)=4x^3$ and $g(4-x)+g(x)=0$, then the value of $\int_{-4}^{4} f\left(x^2\right)dx$ is ....

Q82: Let $f(x)$ and $g(x)$ be two functions satisfying $f\left(x^2\right)+g(4-x)=4x^3$ and $g(4-x)+g(x)=0$, then the value of $\int_{-4}^{4} f\left(x^2\right)dx$ is ....

512.00

Solution:

$$I=2\int_{0}^{4} f\left(x^2\right)dx \text{ \{Even function\}}$$
$$=2\int_{0}^{4}\left(4x^3-g(4-x)\right)dx$$
$$=2\left(\left.\frac{4x^4}{4}\right|_0^4-\int_{0}^{4} g(4-x)\,dx\right)$$
$$=2(256-0)$$
$$=512$$

Q83: The number of times the digit 3 will be written when listing the integers from 1 to 1000 is

....

Q83: The number of times the digit 3 will be written when listing the integers from 1 to 1000 is ....

300.00

Solution:

$3\_\_ = 10 \times 10 = 100$
$\_3\_ = 10 \times 10 = 100$
$\_\_3 = 10 \times 10 = 100$
So, total number of times is $100 + 100 + 100 = 300$

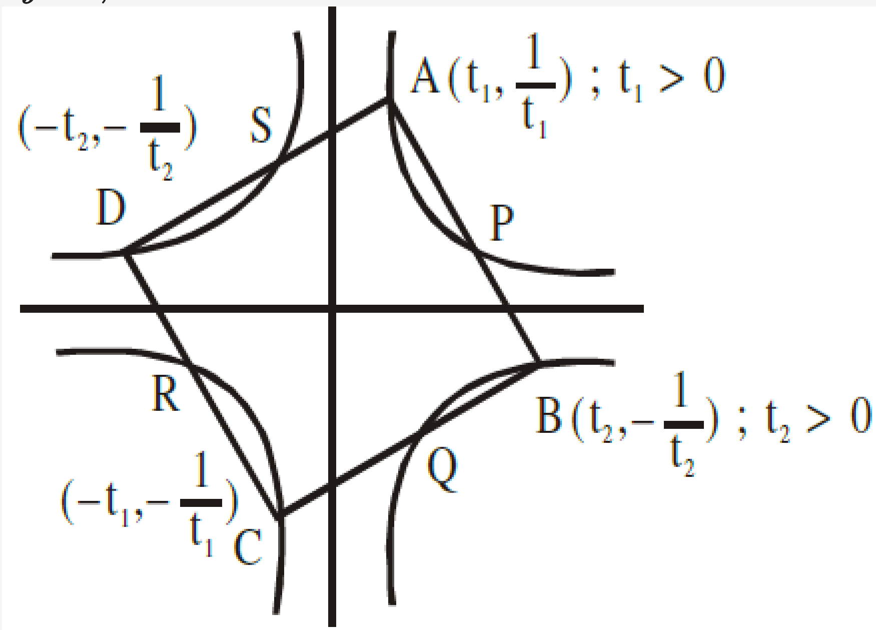
Q84: A square ABCD has all its vertices on the curve $x^2y^2 = 1$. The midpoints of its sides also lie on the same curve. Then, the square of area of ABCD is ....

Q84: A square ABCD has all its vertices on the curve $x^2y^2 = 1$. The midpoints of its sides also lie on the same curve. Then, the square of area of ABCD is ....

80.00

Solution:

$xy = 1, \ -1$



$\frac{t_1+t_2}{2} \cdot \frac{\frac{1}{t_1}-\frac{1}{t_2}}{2} = 1$

$\Rightarrow \ t_2^2 - t_1^2 = 4t_1t_2$

$\frac{1}{t_1^2} \times \left(-\frac{1}{t_2^2}\right) = -1 \ \Rightarrow \ t_1t_2 = 1$

$t_2^2 - t_1^2 = 4$

$\Rightarrow \ t_1^2 + t_2^2 = \sqrt{4^2+4} = 2\sqrt{5}$

$\Rightarrow \ t_2^2 = 2 + \sqrt{5} \ \Rightarrow \ \frac{1}{t_2^2} = \sqrt{5} - 2$

Area $= AB^2 = (t_1 - t_2)^2 + \left(\frac{1}{t_1} + \frac{1}{t_2}\right)^2$

$= 2\left(t_2^2 + \frac{1}{t_2^2}\right) = 4\sqrt{5} \;\Rightarrow\; \text{Area}^2 = 80$

Q85: If $f(x) = \int \frac{5x^8 + 7x^6}{\left(x^2+1+2x^7\right)^2}\, dx,\ (x \geqslant 0)$, $f(0) = 0$ and $f(1) = \frac{1}{K}$, then the value of K is ....

Q85: If $f(x)=\int \frac{5x^8+7x^6}{\left(x^2+1+2x^7\right)^2}\,dx,\ (x \geqslant 0), f(0)=0$ and $f(1)=\frac{1}{K}$, then the value of K is ....

4.00

Solution:

$f(x)=\int \frac{\left(5x^8+7x^6\right)}{x^{14}\left(x^{-5}+x^{-7}+2\right)^2}dx$

Let $x^{-5}+x^{-7}+2=t$

$\left(-5x^{-6}-7x^{-8}\right)dx=dt$

$\Rightarrow\ f(x)=\int-\frac{dt}{t^2}=\frac{1}{t}+c$

$f(x)=\frac{x^7}{x^2+1+2x^7}$

$f(1)=\frac{1}{4}$

Q86: The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is ....

Q86: The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is ....

35.00

Solution:

$\frac{\sum x_i}{25} = 40$ and $\frac{\sum x_i - 60 + N}{25} = 39$

Let age of newly appointed teacher is N

$\Rightarrow \quad 1000 - 60 + N = 975$

$\Rightarrow \quad N = 35$ Years

Q87: Let $z_1,\ z_2$ be the roots of the equation $z^2 +\ az\ +\ 12 =\ 0$ and $z_1,\ z_2$ form an equilateral triangle with origin. Then, the value of $|a|$ is ....

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6.00

Solution:

If $0, \ z_1, \ z_2$ are vertices of equilateral triangles

$\Rightarrow \ 0^2 + z_1^2 + z_2^2 = 0\left(z_1 + z_2\right) + z_1 z_2$

$\Rightarrow \ \left(z_1 + z_2\right)^2 = 3z_1 z_2$

$\Rightarrow \ a^2 = 3 \times 12$

$\Rightarrow \ |a| = 6$

Q88: The number of solutions of the equation $\left|\cot\ x\right| = \cot\ x + \frac{1}{\sin\ x}$ in the interval $\left[0,\ 2\pi\right]$ is ....

Q88: The number of solutions of the equation $\left|\cot\ x\right| = \cot\ x + \frac{1}{\sin\ x}$ in the interval $[0,\ 2\pi]$ is ....

1.00

Solution:

If $\cot x > 0 \ \Rightarrow\ \frac{1}{\sin x} = 0$ (not possible)

If $\cot x < 0 \ \Rightarrow\ 2\cot x + \frac{1}{\sin x} = 0$

$\Rightarrow\ 2\cos x = -1$

$\Rightarrow\ x = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$ (reject)

Q89: The equation of the planes parallel to the plane $x-2y+2z-3=0$ which are at unit distance from the point (1, 2, 3) is $ax+by+cz+d=0$. If $(b-d)=K\,(c-a)$, then the positive value of K is ....

Q89: The equation of the planes parallel to the plane $x - 2y + 2z - 3 = 0$ which are at unit distance from the point (1, 2, 3) is $ax + by + cz + d = 0$. If $(b - d) = K(c - a)$, then the positive value of K is ....

4.00

Solution:

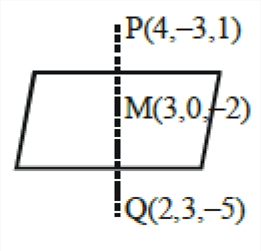
Let plane is $x - 2y + 2z + \lambda = 0$
Distance from $(1,\ 2,\ 3) = 1$
$\Rightarrow\ \frac{|\lambda+3|}{3} = 1\ \Rightarrow\ \lambda = 0,\ -6$
$\Rightarrow\ a = 1,\ b = -2,\ c = 2,\ d = -6\ \text{or}\ 0$
$b - d = 4\ \text{or}\ -2,\ c - a = 1$
$\Rightarrow\ k = 4\ \text{or}\ -2$

Q90: Let the plane $ax + by + cz + d = 0$ bisect the line joining the points (4, –3, 1) and (2, 3, –5) at the right angles. If $a, \ b, \ c, \ d$ are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is ....

Q90: Let the plane $ax + by + cz + d = 0$ bisect the line joining the points (4, –3, 1) and (2, 3, –5) at the right angles. If $a,\ b,\ c,\ d$ are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is ....

28.00

Solution:



Plane is $1\,(x-3) - 3\,(y-0) + 3\,(z+2) = 0$

$x - 3y + 3z + 3 = 0$

$(a^2 + b^2 + c^2 + d^2)_{\min} = 28$